Highlights

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

Yihui Ding, Xiwen Ma, Wenlei Peng, Fangzheng Qi, Jingsong Yang

- Dual-mode SUVs track surface winds and sample subsurface environments.
- Event-specific ERA5–HYCOM fields cover 500 historical TCs.
- Parameter-sharing IPPO scales TC tracking across 4–16 SUVs.
- Current and future SUV configurations are compared at both stages.

Multi-Agent Reinforcement Learning for Surface–Subsurface Observation of Tropical Cyclones Using Submersible Unmanned Vessels

Yihui Ding[a,b,1], Xiwen Ma[a,b,c,1], Wenlei Peng[a,b], Fangzheng Qi[a,b] and Jingsong Yang[a,b,d,*]

[a]School of Oceanography, Shanghai Jiao Tong University, Shanghai, 200030, China
[b]State Key Laboratory of Satellite Ocean Environment Dynamics, Second Institute of Oceanography, Ministry of Natural Resources, Hangzhou, 310012, China
[c]Institute of Automation Chinese Academy of Sciences, Beijing, 100190, China
[d]Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai, 519082, China

ARTICLE INFO

Keywords:
tropical cyclone observation
cooperative task
submersible unmanned vessel
multi-agent reinforcement learning
surface–subsurface observation

ABSTRACT

Tropical cyclones are destructive, while intensity and rapid-intensification forecasts remain limited by sparse near-core observations. Conventional platforms rarely combine rapid transit with surface-wind and subsurface-current sensing. Blue Whale-class submersible unmanned vessels can transit rapidly at the surface to TC cores, then dive to observe temperature, salinity, currents, and biogeochemical properties. We therefore use simulation-based multi-agent reinforcement learning to enable autonomous navigation and environmental observation. The training and evaluation environments combine ERA5 10 m winds and HYCOM currents from 0 to 60 m with IBTrACS tracks for 500 western North Pacific TCs. A parameter-sharing independent proximal policy optimization (IPPO) policy maps shared surface-wind observations or local current histories to continuous control commands at 10 min intervals. The surface policy assumes idealized global sharing of observed wind information, whereas the underwater policy uses no inter-vehicle communication. We compare 12 m/s surface and 1.5 m/s submerged limits, close to current Blue Whale design speeds, with 20 m/s and 2.0 m/s as future upgrades. At the surface, the fleet follows the TC toward high-wind regions despite uncertain deployment, reducing final tracking error and improving observation benefit. Underwater, active control outperforms random control and passive drift in mean experienced current, trajectory extent, and spatial coverage. Compared with four traditional baselines, IPPO achieves better surface tracking distances. Underwater, potential field has the highest mean experienced current, A* the shortest path, and IPPO the largest current-weighted coverage. The framework provides a learning-based closed-loop methodology for future coordinated surface–subsurface TC observation with SUVs.

1. Introduction

Tropical cyclones (TCs) threaten coastal communities, infrastructure, and offshore operations. Climate assessments indicate that some TC changes are detectable despite observational uncertainty [1] and project an increasing proportion of very intense TCs under continued warming [2], while global TC frequency may decline [3] and rapid-intensification events have recently increased in offshore regions [4]. Landfalling TCs over the western North Pacific and China have also become more destructive, with increasing nearshore intensification and exposure [5, 6, 7, 8]. Track forecasts have improved substantially, but intensity prediction remains more difficult: recent intensity guidance has reduced errors [9], yet official forecasts in the western North Pacific continue to carry substantial uncertainty [10]. A central reason is that intensity depends on air–sea enthalpy and momentum exchange, upper-ocean heat content, barrier layers, mesoscale eddies, and TC-induced currents [11, 12].

Existing observing platforms each cover only part of this problem. Spaceborne microwave sensors provide broad but complementary and incomplete views of the near-core wind field. Active scatterometers infer surface wind from ocean backscatter, but lose sensitivity at high wind speeds and can be degraded by rain-induced attenuation and scattering [13]. Passive radiometers infer wind from ocean emissivity and can penetrate some cloud cover, but cloud liquid water, precipitation emission, and atmospheric attenuation contaminate the brightness-temperature signal [13, 14]. Neither sensor class resolves the subsurface ocean [15]. Aircraft obtain high-quality near-core measurements but are expensive and campaign-limited [16, 17], whereas gliders remain at sea for long periods but move much more slowly than a translating TC [18]. AUVs can adapt their sampling to changing environmental and vehicle states [19]. Upper-ocean observations can improve initial states and, in some cases, intensity forecasts [20, 21, 22, 23, 24], but their value is highly sensitive to sampling location and background structure. Targeted observation studies likewise show that forecast benefit depends on sensitive regions, the background state, and the sampling design [25, 26, 27]. These limitations leave near-core wind and upper-ocean states poorly constrained in many forecast initial conditions. Adaptive TC observation is therefore a sequential decision problem involving motion, sampling time, energy, and coordination rather than coverage alone.

*Corresponding author: Jingsong Yang. Email: jsyang@sio.org.cn
ORCID(s): 0009-0009-4133-1571 (Y. Ding); 0000-0001-9212-3523 (Xiwen Ma); 0009-0008-5974-7447 (Fangzheng Qi); 0000-0002-7514-3212 (Jingsong Yang)
[1]These authors contributed equally to this work.

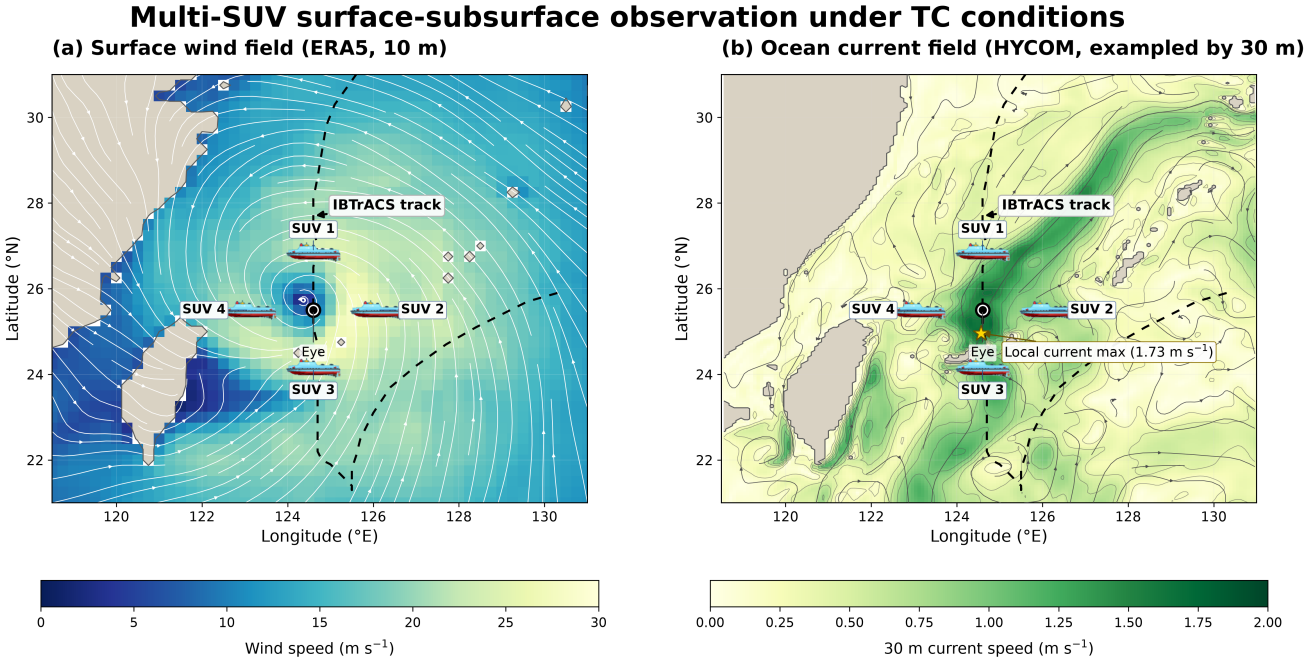


Figure 1: Coordinated surface–subsurface TC observation for a representative HINNAMNOR (2022) snapshot. The left panel shows four SUVs tracking the IBTrACS best track while sampling ERA5 winds at 10 m above the sea surface. The right panel shows the same formation sampling HYCOM currents at 30 m below the sea surface; this depth is shown as an illustrative example, whereas the complete experiments cover the seven depths from 0 to 60 m. The four-vehicle configuration is illustrative, with equal angular spacing; subsequent experiments evaluate fleets of 8, 12, and 16 vehicles.

A submersible unmanned vessel (SUV) combines rapid surface transit with subsurface sensing [28, 29, 30, 31]. The Blue Whale is used as an engineering reference: its public specifications indicate a surface speed of about 23 kn (11.83 m/s), a submerged speed of about 3 kn (1.54 m/s), and an observational payload that includes an acoustic Doppler current profiler and a CTD sensor, with a public reference depth of approximately 60 m [28]. This study uses the current Blue Whale's surface–subsurface motion capabilities as the reference parameterization and evaluates a possible future-upgrade configuration as a comparison case. The vehicles are simplified as point masses for path planning. In the proposed mission, the surface fleet observes 10 m winds, follows the TC toward the high-wind region as closely as possible, and is evaluated by its final tracking error. At depth, four SUVs are assigned to each prescribed layer and seek the strongest observable current. Figure 1 illustrates this surface-to-subsurface task switch; four vehicles are shown at the surface as an example, and the same parameter-sharing policy is later evaluated with 4, 8, 12, and 16 SUVs.

Traditional graph search, sampling-based planning, potential fields, and model predictive control are effective when the cost map and dynamics are known [32, 33, 34, 35, 36, 37]. In a TC, however, the fields are nonstationary and only locally observed, while propulsion is superimposed on ambient flow. Conventional pipelines also estimate a field or waypoint sequence before execution. By contrast, reinforcement learning (RL) maps each new measurement directly to the next action, closing the observation–decision–motion loop. RL has demonstrated strong sequential decision making across diverse domains [38, 39, 40, 41, 42, 43, 44, 45], and independent PPO is competitive in cooperative multi-agent tasks [46, 47, 48]. Multi-AUV adaptive sampling and target tracking provide the ocean-robotics foundations [49, 50, 51, 52, 53].

Three gaps remain. First, prior AUV-based TC studies have focused primarily on center tracking or near-surface observation planning rather than coordinated surface–subsurface observation driven by event-specific reanalysis fields [54, 55]. Second, track-forecast errors require rapid, adaptive transit from an uncertain deployment point to the high-wind region [25, 26, 27]. Third, the closest multi-AUV active-observation study did not test fleet-size scaling [55]. We therefore formulate stochastic policy optimization under partial observability [56, 46, 47], extend prior AUV center-tracking and near-surface active-observation studies [54, 55] to event-specific surface–subsurface observation, and evaluate 4-, 8-, 12-, and 16-SUV fleets.

The main contributions of this study are summarized as follows.

1. We construct an event-specific learning environment for 500 TCs by aligning ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks.
2. Using globally shared wind-speed observations, parameter-sharing IPPO learns inward motion from training-time center rewards and lowers the final

tracking error, measuring how closely the surface fleet follows the TC and enters the high-wind region.
3. We compare active IPPO, random control, and passive drift for subsurface observation with four SUVs at each depth, evaluating mean experienced current speed, trajectory length, and spatial coverage as complementary observation objectives.
4. We compare MARL with four causal traditional planners: an artificial potential-field planner, a receding-horizon A* planner, a lawnmower coverage planner, and an information-greedy planner.

2. Problem Formulation

2.1. Observation Task and Design Requirements

The objective is to coordinate a fleet of submersible unmanned vessels (SUVs) to observe a moving tropical cyclone (TC) across the sea surface and upper ocean. At the surface, each SUV navigates toward the predicted TC center and high-wind region, with the final tracking error used to evaluate how closely the fleet follows the TC. In the operational concept, the fleet would descend after approaching a predefined high-wind observation window. The learning experiments evaluate this mission as two separate episodes: a surface approach episode and a subsurface sampling episode in which four SUVs seek the highest attainable current speed at each prescribed depth. The framework is designed to satisfy four operational requirements:

1. rapid correction of initial deployment errors using surface propulsion;
2. decentralized tracking of the moving TC using onboard wind observations and inter-vehicle observation sharing;
3. subsurface observation at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface;
4. safe operation under action-smoothness and coastline constraints.

The surface maximum speed is varied between 12 and 20 m/s, while the submerged maximum propulsion speed is 1.5 or 2.0 m/s. This difference creates an asymmetric decision problem: surface propulsion is used to follow the TC and reach the high-wind region as closely as possible, quantified primarily by the final tracking error. At depth, the primary learning objective changes to finding and sampling higher-current water, while trajectory length and spatial coverage characterize how broadly the fleet explores the dynamically active region. The four SUVs must therefore weigh active propulsion against passive advection by the ambient current.

2.2. Problem Definition

The fleet contains N SUVs, where $N \in \{4, 8, 12, 16\}$, indexed by $i \in \{1, 2, \dots, N\}$. The position, propulsion velocity, and operating mode of SUV i at time t are denoted by $\mathbf{x}_i(t)$, $\mathbf{p}_i(t)$, and $m_i(t) \in \{\mathrm{s}, \mathrm{u}\}$, respectively, where s and u denote the surface and subsurface modes. Horizontal positions and reported distances are expressed in km, whereas propulsion and ocean-current velocities are expressed in m/s. The propulsion speed satisfies

$$\|\mathbf{p}_i(t)\| \le \begin{cases} V^{\mathrm{s}}_{\max}, & m_i(t) = \mathrm{s}, \\ V^{\mathrm{u}}_{\max}, & m_i(t) = \mathrm{u}. \end{cases} \tag{1}$$

Each simulated episode is assigned to one stage, consistent with the separate surface and subsurface environments used in training and evaluation. Thus, $m_i(t)$ selects the applicable speed bound and motion law for that episode; continuous diving, ballast transitions, and resurfacing are not simulated within a single decision trajectory.

For the motion equations, the km/min representations of propulsion and ocean-current velocities are denoted by $\mathbf{v}^{\mathrm{p}}_i$ and $\mathbf{v}_{\mathrm{ocean}}$; the device-level speed bounds above remain reported in m/s. The horizontal motion in km is

$$\dot{\mathbf{x}}_i(t) = \begin{cases} \mathbf{v}^{\mathrm{p}}_i(t), & m_i(t) = \mathrm{s}, \\ \mathbf{v}^{\mathrm{p}}_i(t) + \mathbf{v}_{\mathrm{ocean}}\left(\mathbf{x}_i(t), z_i, t\right), & m_i(t) = \mathrm{u}, \end{cases} \tag{2}$$

where $z_i \in \{0, 10, 20, 30, 40, 50, 60\}$ m. Time t is measured in minutes, and $\mathbf{v}_{\mathrm{ocean}}(\cdot)$ is the HYCOM current at the corresponding position and depth. Thus, the underwater ground velocity is the superposition of active propulsion and background advection.

For numerical implementation, the continuous model is integrated explicitly at the decision interval $\Delta t = 10$ min:

$$\mathbf{x}_{i,k+1} = \mathbf{x}_{i,k} + \Delta t \begin{cases} \mathbf{v}^{\mathrm{p}}_{i,k}, & m_{i,k} = \mathrm{s}, \\ \mathbf{v}^{\mathrm{p}}_{i,k} + \mathbf{v}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k), & m_{i,k} = \mathrm{u}. \end{cases} \tag{3}$$

Here $\Delta t = 10$ min. The current field is evaluated at the beginning of each interval and interpolated to the current vehicle position. For the active underwater reward, we write this pre-move current as $\mathbf{u}_{i,k} = \mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,k}, z_i, t_k)$.

This equation is deliberately a reduced-order horizontal point-mass kinematic–advection model rather than a six-degree-of-freedom rigid-body dynamic model. It does not resolve added mass, hydrodynamic drag, roll, pitch, yaw or heading dynamics, buoyancy and ballast dynamics, thruster transients, wave loads, or actuator delay. This simplification is consistent with the information content of the environmental forcing. ERA5 winds are available at 0.25° and hourly resolution, whereas HYCOM currents are available at approximately 0.04° by 0.08° and three-hourly resolution. These fields are interpolated to the 10 min decision interval and to each vehicle position, but such interpolation does not recover high-frequency wave forcing, local wake effects, or unobserved vehicle attitude and actuator states. Introducing a detailed six-degree-of-freedom model would therefore require platform-specific hydrodynamic coefficients and forcing information that are not commensurate with the present data, and could imply a level of numerical precision unsupported by the inputs. The reduced model retains the first-order effects relevant to mission-level planning: mode-dependent speed limits, propulsion commands, ambient-current advection in subsurface mode, and

coastal constraints. Accordingly, the results are interpreted as mission-planning and coordination evidence rather than as predictions of attitude, structural loading, or low-level controller response. Coupling this planning layer to an experimentally calibrated six-degree-of-freedom vehicle model is left for future work.

2.3. Surface–Subsurface Two-Stage Task

The SUVs first perform 20 h of surface tracking, using coordinated propulsion to follow the TC and approach its high-wind region. At the end of this stage, once the fleet is close to the TC core, the vehicles dive and begin subsurface sampling. Because the present simulation does not model the dive dynamics, the surface and subsurface stages are represented as two time-aligned episodes. The operational protocol specifies the 20 h surface stage and subsequent descent near the core, whereas the simulation joins the two episodes at the segment boundary.

The local observation of SUV i is

$$\mathbf{o}_i(t) = \begin{cases} \left[\mathbf{s}_i(t),\ \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{s}, \\ \left[\mathbf{h}_i(t),\ \mathbf{v}_i(t)\right], & m_i(t) = \mathrm{u}, \end{cases} \tag{4}$$

where $\mathbf{s}_i(t)$ is the stage-specific measurement block, $\mathbf{v}_i(t)$ is the normalized commanded propulsion velocity $\mathbf{v}_i = \mathbf{p}_i / V_{\max}^{m_i(t)}$, and $\mathbf{h}_i(t)$ is an optional causal history block. In the final surface experiments, $\mathbf{s}_i(t) \in \mathbb{R}^{3N}$ contains the normalized speed and two wind components at 10 m above the sea surface sampled by all N SUVs; the surface policy has no history block. Surface operation assumes idealized, lossless, and instantaneous inter-vehicle communication: every SUV receives the wind observations acquired by all other SUVs in the fleet as part of its policy input. In the underwater active-control comparison, no inter-vehicle communication is assumed. The observation contains no separate instantaneous-current block in addition to the history; instead, define the local current feature $\mathbf{c}_{i,t} = [\|\mathbf{u}_{\mathrm{ocean}}(\mathbf{x}_{i,t}, z_i, t)\|, u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y}]^\top$, where $\mathbf{u}_{\mathrm{ocean}} = (u_{\mathrm{ocean},x}, u_{\mathrm{ocean},y})$, and $\mathbf{h}_i(t) = [\mathbf{c}_{i,t-4}^\top, \ldots, \mathbf{c}_{i,t}^\top]^\top \in \mathbb{R}^{15}$, which is the causal five-frame history of SUV i's current speed and two components at its assigned depth; before five measurements are available, the earliest observed frame is repeated to fill the history. Each underwater SUV therefore knows neither the current observations nor the positions or actions of the other SUVs. The complete reanalysis field, the global best-track center, and future environmental states are not provided to the policy. These two communication assumptions are intentionally different and mark the bounds of the present study. The ideal all-to-all surface link represents a favorable satellite or radio-connected operating mode and is therefore an upper-bound assumption. The no-communication underwater case is the conservative opposite bound: shared policy parameters are loaded separately on each vehicle, but no observation, position, or action is exchanged at runtime. The surface payload is $3N$ scalar wind values per vehicle per decision, whereas the underwater inter-vehicle message payload is zero. The reported experiments do not model bandwidth, packet loss, acoustic latency, or intermittent surfacing. Accordingly, neither the ideal surface link nor the fully decentralized subsurface case is presented as a validated communication system. Accordingly, the surface observation dimension is $d_o^{\mathrm{s}} = 3N + 2$. The underwater five-frame history contains $5 \times 3 = 15$ scalars; adding the two normalized propulsion components gives $d_o^{\mathrm{u}} = 15 + 2 = 17$.

During training, the reward function can access the distance between each SUV and the best-track TC center. This distance supervises the association between local 10 m wind patterns and proximity to the TC center/high-wind region; the center coordinate itself is not an actor input. During validation, rewards do not influence actions. The frozen policy receives only the shared fleet-level wind samples and must track an unseen TC, thereby testing whether the learned wind-to-center relation generalizes.

The normalized action is

$$\mathbf{a}_i(t) \in [-1, 1]^2, \tag{5}$$

and is projected onto the unit disk before being mapped to propulsion in both modes:

$$\tilde{\mathbf{a}}_i(t) = \frac{\mathbf{a}_i(t)}{\max\left(1, \|\mathbf{a}_i(t)\|_2\right)}, \qquad \mathbf{p}_i(t) = V_{\max}^{m_i(t)} \bar{\mathbf{a}}_i(t). \tag{6}$$

This mapping ensures that the total propulsion speed never exceeds the mode-specific speed limit. Here $\bar{\mathbf{a}}_i(t)$ is the executed normalized command after action smoothing, unit-disk projection, and any stage-specific safety projection; without a safety intervention, $\bar{\mathbf{a}}_i(t) = \tilde{\mathbf{a}}_i(t)$.

The cooperative task is formulated as a decentralized partially observable Markov decision process (Dec-POMDP):

$$\mathcal{M} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}, P, R, \{\Omega_i\}, O, \gamma \rangle, \tag{7}$$

where $\mathcal{I}$ is the set of SUVs, $\mathcal{S}$ is the joint environment state, $\mathcal{A}_i$ and Ω_i are the local action and observation spaces of vehicle i, $P(s_{t+1} \mid s_t, \mathbf{a}_t)$ is the transition probability, $R(s_t, \mathbf{a}_t, s_{t+1}) \in \mathbb{R}^N$ is the local-reward vector, and $O(\mathbf{o}_{t+1} \mid s_{t+1}, \mathbf{a}_t)$ is the joint observation function. The shared team reward before clipping is $\bar{r}_t = N^{-1} \sum_{i=1}^{N} r_{i,t}$, where $r_{i,t}$ is the ith element of $R(s_t, \mathbf{a}_t, s_{t+1})$, and γ is the discount factor.

The policy operates as a closed perception–decision–action loop rather than following a precomputed route. At each 10 min interval,

$$\begin{aligned} \mathbf{o}_{i,t} &\longrightarrow \mathbf{a}_{i,t} = \pi_\theta(\mathbf{o}_{i,t}) \\ &\longrightarrow \mathbf{x}_{i,t+1} \longrightarrow \mathbf{o}_{i,t+1}. \end{aligned} \tag{8}$$

Each updated observation therefore becomes the next control input. This feedback structure lets the fleet adapt to changing wind and current fields, partial observability, and deployment uncertainty without a static waypoint sequence.

Table 1
Environmental datasets used to construct the 500-TC observation environments. The dataset integrates ERA5 surface winds, HYCOM currents at seven depths, and IBTrACS best tracks.

Data product	Source	Variables used	Native grid	Native time
Surface wind at 10 m above sea level	ERA5	u_{10}, v_{10}, speed, direction	$0.25° \times 0.25°$	Hourly
Current at 0–60 m ocean depth	HYCOM	water_u, water_v, speed at 0/10/20/30/40/50/60 m	$\sim 0.04°$ lat $\times 0.08°$ lon	3-hourly
Best track	IBTrACS	Position, time, wind, pressure	Point track	Mainly 3/6-hourly

Note: although the SUV platform can carry sensors for observing temperature, salinity, and currents, the reinforcement-learning policy uses only surface wind and subsurface current observations as environmental inputs.

3. Data and Simulation Environment

3.1. Reanalysis and Best-Track Products

ERA5 supplies hourly zonal and meridional wind components at 10 m above the sea surface on a regular 0.25° grid [1] [57]. Wind speed and direction are calculated from u_{10} and v_{10}. HYCOM supplies three-hourly zonal and meridional currents at ocean depths of 0, 10, 20, 30, 40, 50, and 60 m below the sea surface [2] [58]. The native horizontal grid is approximately 0.04° in latitude by 0.08° in longitude. IBTrACS provides TC time, center position, maximum wind, minimum pressure, and related metadata at mostly 3 or 6 h intervals [3] [59]. The dataset-selection domain used to identify candidate systems is 98.7°E–171.6°E and 6.8°N–56.7°N. Individual event fields are stored on event-specific crops, so a track may extend beyond this screening box. The overall event set spans 19 November 2007 to 3 December 2025 UTC. Thus, the surface observation variable is the 10 m wind speed and direction, whereas the subsurface observation variables are the horizontal current velocity, represented by water_u and water_v, and the derived current speed over the 0–60 m layer. Table 1 summarizes the products, variables, native grids, and temporal resolutions used to construct the event-specific environments.

3.2. Spatiotemporal Alignment and Track Segmentation

Hourly ERA5 winds and three-hourly HYCOM currents are linearly interpolated to a common 10 min simulation clock and queried at each SUV position. The environmental trajectories are precomputed and do not respond to vehicle motion; this one-way forcing assumption is appropriate for mission-level planning at the scale of a TC. The policy receives only local reconstructed values at the current time, not future fields or the complete reanalysis map. Invalid interpolated values are replaced by a neutral value, and all policy observations are checked for finiteness. The best-track center is used only for training rewards and evaluation.

Commands are held constant within each decision interval:

$$\mathbf{a}_i(t) = \mathbf{a}_{i,k}, \quad t_k \le t < t_{k+1}, \quad t_{k+1} - t_k = \Delta t = 10\,\text{min}. \tag{9}$$

The environmental fields, TC track, action, and position update therefore share one causal time grid. The 10 min step balances control responsiveness against the hourly and three-hourly resolution of the source products; a shorter step would add samples without adding resolved information. The same interval is used for IPPO and all traditional planners.

Each IBTrACS track is first linearly interpolated to the 10 min simulation interval. The cumulative along-track distance is computed using the local equirectangular approximation

$$\Delta s_k = \sqrt{\left[111\Delta\lambda_k \cos(\varphi_k)\right]^2 + \left(111\Delta\varphi_k\right)^2}, \tag{10}$$

where $\Delta\lambda_k$ and $\Delta\varphi_k$ are longitudinal and latitudinal increments in degrees, and 111 km is the mean meridional arc length per degree. Target segment lengths are sampled uniformly from 300–500 km, with a fixed 250 km stride, 50–250 km overlap, and an average overlap of about 150 km. Residual track shorter than 300 km is discarded. After assignment to 10 min track samples, realized lengths are 292.1–507.1 km (median 396.4 km).

The 500-TC dataset yields 7,248 segments from 495 TCs. The 450-TC training split contributes 6,480 segments from 445 usable TCs, whereas the 50 held-out TCs contribute 768 segments. The split is made at the TC level, so no TC appears in both sets. Segment durations are 2.2–103 h (median 16.5 h, mean 20.0 h). These segments provide mission-scale evaluation windows; battery depletion and a hard energy budget are not simulated. The platform can carry current, CTD, pH, chlorophyll, and turbidity sensors [28], while the control experiments use current velocity as the decision-making observation.

4. Multi-Agent Reinforcement Learning Method

4.1. Blue Whale-Class Platform Modeling

Each SUV is represented as a horizontal point mass with mode-dependent speed bounds and position updates from Eqs. (1)–(2). The surface actor uses globally shared 10 m wind observations and its normalized velocity, whereas the underwater actor uses a causal five-frame history of its

[1] ERA5 download portal: https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels.

[2] HYCOM GLBy0.08 dataset: https://www.hycom.org/data/glby0p08/expt-93pt0; HYCOM ESPC-D-V02 dataset: https://www.hycom.org/data/espcd-v02.

[3] IBTrACS download portal: https://www.ncei.noaa.gov/products/international-best-track-archive.

Table 2
Public Blue Whale reference and the two simulation configurations. The second simulation column is a simple parameterization of the present-day reference platform, whereas the third represents a possible future-upgrade configuration with higher surface and subsurface speeds for comparison. These scenarios are modeling assumptions, not certified vehicle data. The 20 h value is a common simulation horizon, not a platform-endurance claim.

Property	Public reference	Current-platform scenario	Future-upgrade scenario
Role	Engineering reference	Simple current-platform parameterization	Potential future capability
Surface speed	23 kn, ≈11.83 m/s	12 m/s	20 m/s
Submerged speed	3 kn, ≈1.54 m/s	1.5 m/s	2.0 m/s
Submerged endurance (public reference)	≥4 h at 3 kn; ≥72 h standby	–	–
Underwater evaluation horizon	–	20 h (simulation)	20 h (simulation)
Sampling depth	≈60 m	0–60 m (7-layer)	0–60 m (7-layer)
Payload	≥800 kg	800 kg	800 kg
Surface fleet size	Single platform	4/8/12/16	4/8/12/16
Decision interval	Not specified	10 min	10 min
Surface formation	–	100 km ring, $360°/N$	100 km ring, $360°/N$

Table 3
IBTrACS TC translation speeds for the 500-TC dataset. The percentages denote the fraction of main-track samples not exceeding each platform speed. These values describe the speed envelope only; tracking also depends on bearing error and remaining segment duration.

Quantity	P50	P90	P95	P99	Samples	TCs
TC translation speed (m s^{-1})	5.14	9.77	13.38	20.06	30,498	500
Envelope	Samples within speed		Samples above speed		Interpretation	
1.5 m/s submerged	3.6%		96.4%		local sampling, not TC chasing	
2.0 m/s submerged	7.8%		92.2%		local sampling, not TC chasing	
12 m/s surface	93.8%		6.2%		current-platform scenario	
20 m/s surface	98.9%		1.1%		future-upgrade scenario	

local current and normalized velocity. Path length is reported for evaluation but is not an actor input. Surface motion neglects ocean-current advection because 12–20 m/s propulsion dominates the relevant surface transport; underwater motion retains advection because 1.5–2.0 m/s propulsion is comparable to the TC-induced current.

The two simulation configurations compare present and potential future platform capabilities. The 12 m/s surface and 1.5 m/s submerged limits are close to the public Blue Whale design speeds, whereas 20 m/s and 2.0 m/s represent a future-upgrade case. Table 2 reports the complete parameterization and identifies all values that are modeling assumptions rather than certified specifications.

Across 30,498 IBTrACS track samples, the median, 95th-percentile, and 99th-percentile TC translation speeds are 5.14, 13.38, and 20.06 m/s (Table 3). The 12 and 20 m/s surface limits cover 93.8% and 98.9% of these samples, respectively. The submerged limits cover only 3.6% and 7.8%; the subsurface task is therefore local adaptive sampling within the 200 km initialization region, not overtaking and following the TC center.

4.2. Surface and Subsurface Reward Design

The surface objective is to follow the translating TC as closely as possible and reach the high-wind region, with the final tracking error as the primary operational metric. Let $d_{i,k+1}$ denote the post-transition distance from SUV i to the TC center after executing the command at decision step k. The surface reward used in the implementation is

$$\begin{aligned} r_i^{\mathrm{surf}} = & -\frac{\min(d_{i,k+1}, D_{\max})}{D_{\mathrm{scale}}} + b_{\mathrm{near}} \mathbf{1}_{d_{i,k+1} < d_{\mathrm{near}}} \\ & - \frac{W_{\mathrm{form}} \max(0, d_{i,k+1} - R_{\mathrm{form}})}{D_{\mathrm{scale}}} \\ & - \lambda_u \|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2 + r_i^{\mathrm{coast}}, \end{aligned} \tag{11}$$

where the first term is a bounded proximity cost and the second term provides a discrete near-center bonus with $b_{\mathrm{near}} = 1$. Here $D_{\max} = 300$ km caps the proximity cost, $D_{\mathrm{scale}} = 100$ km normalizes the distance terms, and $d_{\mathrm{near}} = 100$ km defines the near-center reward region. The third term adds an additional radius-excess penalty beyond $R_{\mathrm{form}} = 100$ km, and the fourth term penalizes abrupt action changes with $\lambda_u = 0.05$. The term r_i^{coast} represents the coastal-safety reward defined below. The 300 km cap prevents distant

errors from producing an excessively steep gradient and reduces premature policy collapse during early training. After aggregation over SUVs, the team average reward is clipped to $[-5, 2]$.

For subsurface observation, four SUVs are assigned to each prescribed depth. Their primary objective is not to follow the surface TC center directly, but to observe the highest possible current speed. Trajectory length and spatial coverage are then evaluated as complementary measures of observation extent. Low-speed SUVs cannot generally overtake a translating TC, and the vertical projection of the surface center is not necessarily the location of the strongest subsurface current. For the active-control comparison, the current-seeking reward is

$$\begin{aligned} r_i^{\mathrm{sub}} = \; & \min\left(\frac{\|\mathbf{u}_{i,k}\|}{V_{\mathrm{norm}}}, R_{\max}\right) \\ & - \lambda_E \|\tilde{\mathbf{a}}_{i,k}\|_2^2 + \beta_{\parallel} \max(0, \cos\theta_i) \\ & + r_i^{\mathrm{coast}}, \end{aligned} \tag{12}$$

where $V_{\mathrm{norm}} = 1.0$ m/s, $R_{\max} = 2.0$, $\lambda_E = 0.05$, and $\beta_{\parallel} = 0.5$. The angle θ_i denotes the direction difference between propulsion and the pre-move current $\mathbf{u}_{i,k}$, so the positive-alignment term is activated only when the SUV pushes downstream; formally, we define $\cos\theta_i = 0$ when either vector is zero and otherwise set $\cos\theta_i = (\tilde{\mathbf{a}}_{i,k} \cdot \mathbf{u}_{i,k})/(\|\tilde{\mathbf{a}}_{i,k}\|_2 \|\mathbf{u}_{i,k}\|_2)$. The current-intensity reward is therefore evaluated at $\mathbf{x}_{i,k}$ before the state update. For the local maximum-current objective, passive drift is retained as the no-propulsion reference because it follows the ambient flow directly and does not depend on a learned reward policy.

4.3. Coastal Safety and Action Smoothing

The land mask is converted into a nonnegative offshore-distance field. A vehicle inside land receives a large penalty, and a vehicle within the buffer distance D_{buf} receives

$$r_i^{\mathrm{coast}} = \begin{cases} -P_{\mathrm{land}}, & D_i = 0, \\ -W_b \max(0, 1 - D_i/D_{\mathrm{buf}}), & 0 < D_i < D_{\mathrm{buf}}, \\ 0, & D_i \geq D_{\mathrm{buf}}, \end{cases} \tag{13}$$

where $D_i = D(\mathbf{x}_{i,k+1})$ is the post-transition offshore distance, with $D_i = 0$ on land, and P_{land} and W_b are positive penalty coefficients.

For the surface stage, three complementary mechanisms reduce coastline intrusion: offshore-only rejection sampling during deployment, a soft nearshore penalty, and an action-level safety filter. The main surface experiments use a minimum deployment distance of 100 km, a 100 km nearshore buffer, and a 100 km safety margin. Best-of-rejection deployment favors offshore initial states, the reward discourages trajectories near land, and the safety filter removes the onshore action component when an SUV approaches the shoreline. For the subsurface stage, the nearshore penalty is retained and an episode is truncated at landfall, but offshore-only initialization and action-level safety filtering are not applied. These mechanisms reduce land exposure in the surface results but are not posed as an unconditional hard constraint on every vehicle position.

To reduce oscillatory control, the main surface experiments use an exponential action filter:

$$\mathbf{a}_k^{\mathrm{filt}} = (1 - \beta)\mathbf{a}_k^{\mathrm{raw}} + \beta \mathbf{a}_{k-1}^{\mathrm{filt}}, \tag{14}$$

where $\beta \in [0, 1)$ controls the degree of temporal smoothing. The filter uses only the current and previous actions and therefore does not introduce future-information leakage. The filtered command $\mathbf{a}_k^{\mathrm{filt}}$ is then passed through the unit-disk projection and speed mapping in Eq. (6).

4.4. Parameter-Sharing Independent PPO

All SUVs contribute to the same observation and tracking objective, and the training return is defined as the team-average reward. The problem is therefore a fully cooperative task, rather than a competitive or mixed cooperative–competitive task. This cooperative reward structure motivates the use of IPPO, in which decentralized vehicle actions are optimized against a common team objective while policy parameters are shared across vehicles.

In this paper, IPPO denotes the cooperative and parameter-sharing implementation of independent PPO. All SUVs use the same actor and critic weights, while their observations and action samples remain vehicle-specific. Actions are sampled independently, and the per-agent values are summed for training without a global-state centralized critic. The main surface model uses a memoryless multilayer perceptron, whereas the active underwater controller uses a five-frame current history. The shared cooperative return is the team-average reward, with value estimate $V_\phi(\mathbf{o}_t) = \sum_{i \in \mathcal{I}} v_\phi(\mathbf{o}_{i,t})$, and a single clipped PPO objective is optimized over the factorized multi-agent action vector:

$$\begin{aligned} L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(& \rho_t(\theta)\hat{A}_t, \\ & \mathrm{clip}\left(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon\right)\hat{A}_t \Big)\Big], \end{aligned} \tag{15}$$

where

$$\rho_t(\theta) = \prod_{i \in \mathcal{I}} \frac{\pi_\theta(a_{i,t} \mid o_{i,t})}{\pi_{\theta_{\mathrm{old}}}(a_{i,t} \mid o_{i,t})}, \tag{16}$$

and $\hat{A}_t$ is estimated by generalized advantage estimation from the team-average reward. The actor objective is complemented by a clipped value loss and an entropy bonus. During execution, each SUV computes its action from its own observation and the shared policy network.

To compare reinforcement-learning algorithm choices, IPPO, SAC, and TD3 were trained with two seeds and the same 2×10^6 environment steps on representative surface and subsurface cases. IPPO achieved the lowest surface final error (317.3 km versus 494.7 and 548.5 km) and the strongest combined underwater current-exposure and coverage result, while requiring less training time and no replay buffer. We therefore use IPPO for the main, formation, and planner

Algorithm 1 Training and deployment of parameter-sharing IPPO

Input: Training TC split, stage-specific environment, fleet size N, speed limit $V_{\max}$, reward coefficients, and random seed
Output: Shared actor and critic parameters

Initialize the actor, critic, and parallel environment workers
for training iteration $m = 1, \dots, M$ **do**
  Reset the stage-specific environment and obtain $\mathbf{o}_{i,t}$ for all $i = 1, \dots, N$
  for rollout step t **do**
    Sample or select an action for each SUV from $\pi_\theta(\cdot \mid \mathbf{o}_{i,t})$
    Apply the stage-specific action filter, project each agent's action onto the unit disk, and apply the safety filter
    Advance all SUVs and collect the team-average reward
  end for
  Compute GAE advantages and returns for the team-average reward
  Update the shared actor and critic with the clipped PPO objective
end for
Freeze the actor and load the same parameters on every SUV
for each deployment episode **do**
  Observe local or shared data according to the stage communication model
  Compute $\mathbf{a}_{i,t}$ from the shared actor
  Apply action smoothing, circular projection, and safety filtering
  Execute the action and observe the next state
end for

comparisons; the two-seed result is treated as bounded evidence rather than a universal algorithm ranking.

The complete reward coefficients, safety parameters, network dimensions, and PPO hyperparameters are summarized in Appendix A. The project repository and software package sources are listed in the Code and package availability section.

4.5. Surface and Subsurface Evaluation

The center-distance reward is used only during training and is absent from validation; validation therefore tests whether a frozen policy can infer the TC direction from wind observations in unseen events. The primary surface metric is the final fleet-averaged SUV-to-TC-center distance. A smaller value indicates closer core entry and higher observation benefit. Time-averaged distance is reported as a secondary process diagnostic.

For training seed s and its validation dataset, the seed-level value is

$$D_{\text{final}}^{(s)} = \frac{1}{M_s} \sum_{e=1}^{M_s} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_{i,e}(T_e) - \mathbf{x}_{\text{TC},e}(T_e) \right\|_2 , \quad (17)$$

where T_e is the final decision time of validation segment e, and M_s is the number of validation segments. Results are averaged over ten training seeds and reported with the sample standard deviation.

The process-level fleet-mean time-averaged distance for the same seed is

$$D_{\text{avg}}^{(s)} = \frac{1}{M_s} \sum_{e=1}^{M_s} \frac{1}{K_e} \sum_{k=1}^{K_e} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_{i,e,k} - \mathbf{x}_{\text{TC},e,k} \right\|_2 , \quad (18)$$

where K_e indexes the post-transition decision intervals. Distance is averaged over SUVs, episode time, and validation segments, then over training seeds.

Surface evaluation covers 4/8/12/16 SUVs on a 100 km ring with equal angular spacing. Training samples formation-center offsets within an 800 km radius; the frozen policies are evaluated at 100, 200, 400, 600, and 800 km. These are mission-scale test conditions, not tuned deployment optima. Metrics are first averaged over validation segments within each seed, then summarized over ten seeds. Paired 12-versus-20 m/s tests are two-sided and uncorrected across the twenty fleet-size/radius comparisons, so their p-values are exploratory.

For the subsurface evaluation, four SUVs at each depth are tested over 20 h at 0, 10, 20, 30, 40, 50, and 60 m under 1.5 and 2.0 m/s speed limits. The subsurface episode starts from a position sampled within 200 km of the TC center at the corresponding surface-segment end time; the surface endpoint itself is not transferred. There is no inter-vehicle communication, so each SUV uses only its own five-frame current history and propulsion state. Active IPPO, random control, and passive drift share the same episodes, deployment seeds, motion model, and speed bound.

The three strategies differ only in how propulsion is generated:

$$\begin{aligned} \mathbf{a}_i^{\text{active}} &= \text{clip}\left(\pi_\theta(\mathbf{o}_i), -1, 1\right), \\ \mathbf{a}_i^{\text{random}} &\sim \mathcal{U}\left([-1, 1]^2\right), \\ \mathbf{a}_i^{\text{drift}} &= \mathbf{0}. \end{aligned} \quad (19)$$

During evaluation, the active IPPO policy uses its deterministic mean action, whereas random control resamples a two-dimensional action every 10 min. Active IPPO and random control share the same propulsion limit and equations of motion; only the active policy conditions its action on the local current history. Passive drift applies zero propulsion and advances solely by ambient current advection.

For evaluation reporting, let K be the number of recorded position increments and let $k = 0, \dots, K$ index the archived positions. The recorded ground-path length, mean experienced current speed, and nominal coverage are

$$L_i(z) = \sum_{k=0}^{K-1} \left\| \mathbf{x}_{i,k+1} - \mathbf{x}_{i,k} \right\|_2 , \tag{20}$$

$$\bar{V}_i(z) = \frac{1}{K+1} \sum_{k=0}^{K} \left\| \mathbf{u}_{\text{ocean}}(\mathbf{x}_{i,k}, z, t_k) \right\|_2 . \tag{21}$$

$$C(z) = A_{\text{cell}} \left| \bigcup_{i=1}^{N} \bigcup_{k=0}^{K} \mathcal{G}(\mathbf{x}_{i,k}) \right| , \tag{22}$$

where $\mathcal{G}(\mathbf{x})$ is the 0.05° grid cell containing position $\mathbf{x}$, $|\cdot|$ counts unique cells, and $A_{\text{cell}} \approx (0.05° \times 111 \text{ km})^2$ is the nominal grid-cell area used for cross-method comparison. The figures report the fleet means $N^{-1} \sum_{i=1}^{N} L_i(z)$, $N^{-1} \sum_{i=1}^{N} \bar{V}_i(z)$, and the nominal coverage $C(z)$. Because neither current intensity nor coverage alone captures observation value, we also report the episode-level current-weighted coverage

$$C_{\text{w}}(z) = \frac{1}{M} \sum_{e=1}^{M} \left[\frac{1}{N} \sum_{i=1}^{N} \bar{V}_{i,e}(z) \right] C_e(z), \tag{23}$$

where M is the number of paired validation episodes. This quantity assigns a higher value to covering a broad region while encountering a strong current within that region, without assuming that either objective dominates in every episode.

4.6. Traditional Planning Baselines

We include four non-learned planners as secondary references [33, 35, 37]. The artificial-potential-field planner moves along the local gradient of observed wind or current speed and adds an offshore repulsion inside the coastal buffer. The receding-horizon A* planner builds an online local value map, re-plans a three-step minimum-cost route, executes the first action, and refreshes the plan every 15 decisions. The boustrophedon lawnmower planner follows fixed parallel coverage legs with phase offsets between vehicles and uses no environmental value map. The information-greedy planner evaluates eight candidate unit actions and selects the one with the largest predicted local wind or current value plus the coastal preference.

None of the planners receives the TC center, the complete reanalysis field, future frames, or the learned policy state. Surface planners use the same shared wind samples and fleet navigation state as IPPO. Underwater, each vehicle maintains a private map because communication is disabled. All four use the same 10 min interval, speed envelope, circular projection, surface safety filter, and subsurface landfall termination. They require no training and therefore have no training-seed variance. Surface planners are evaluated on all 768 held-out segments, whereas underwater planners use the same 500 paired episodes per depth-speed setting (50 test TCs with ten deployment seeds).

5. Simulation and Results

5.1. Surface Active Tracking

Using the evaluation protocol in Section 4.5, we evaluate the extent to which the trained policies reduce forecast-induced deployment errors and approach the moving TC center and high-wind region.

Figure 3 reports the primary final tracking error from Eq. (17), for which smaller values are better, and the secondary time-averaged distance. Each value summarizes the validation dataset over ten independent training seeds.

Using the final tracking error as the operational observation benefit, the right panel of Fig. 3 reports the primary speed-limit comparison. Averaged over the four fleet sizes and ten training seeds, the 20 m/s policy obtains final tracking errors of 197.1, 209.2, 260.9, 362.1, and 524.0 km at deployment radii of 100, 200, 400, 600, and 800 km, respectively. The corresponding 12 m/s values are 221.0, 238.9, 317.5, 450.9, and 635.1 km. Thus, 20 m/s yields a lower final tracking error in all 20 matched fleet-size/radius comparisons, corresponding to a higher operational observation benefit. Its fleet-averaged advantage increases from 23.8 km at 100 km to 111.1 km at 800 km.

The absolute distances still increase with deployment radius for both speed limits, which is expected because a finite propulsion speed cannot eliminate a larger initial displacement within every segment duration. The objective is therefore to minimize the final tracking error at each error level rather than to make it independent of deployment error. The final-tracking-error advantage of 20 m/s is larger than the corresponding time-averaged advantage: 23.8 versus 10.1 km at 100 km and 111.1 versus 84.2 km at 800 km. This indicates that the higher-speed policy continues correcting residual separation late in the episode rather than merely remaining closer during the initial transit.

This benefit involves a clear propulsion trade-off. Relative to 20 m/s, the 12 m/s policy reduces path length per SUV by 78.7, 95.5, 147.4, 196.6, and 233.3 km at the five radii and lowers mean ground speed by 1.74–4.18 m/s. Accordingly, 20 m/s provides the higher final observation benefit, whereas 12 m/s is preferable when shorter trajectories and lower speed are prioritized.

Time-averaged distance provides a secondary process-level diagnostic of whether proximity is maintained throughout an episode. Across the twenty matched fleet-size/radius comparisons, the 20 m/s policy has the smaller mean time-averaged distance in all 20 cases. The median paired difference ($d_{12} - d_{20}$) is +51.8 km. Uncorrected two-sided paired tests indicate lower distance at 20 m/s in 19 cases and at 12 m/s in none of the cases at $p < 0.05$. Because these comparisons were not adjusted for multiplicity, they provide exploratory rather than confirmatory evidence.

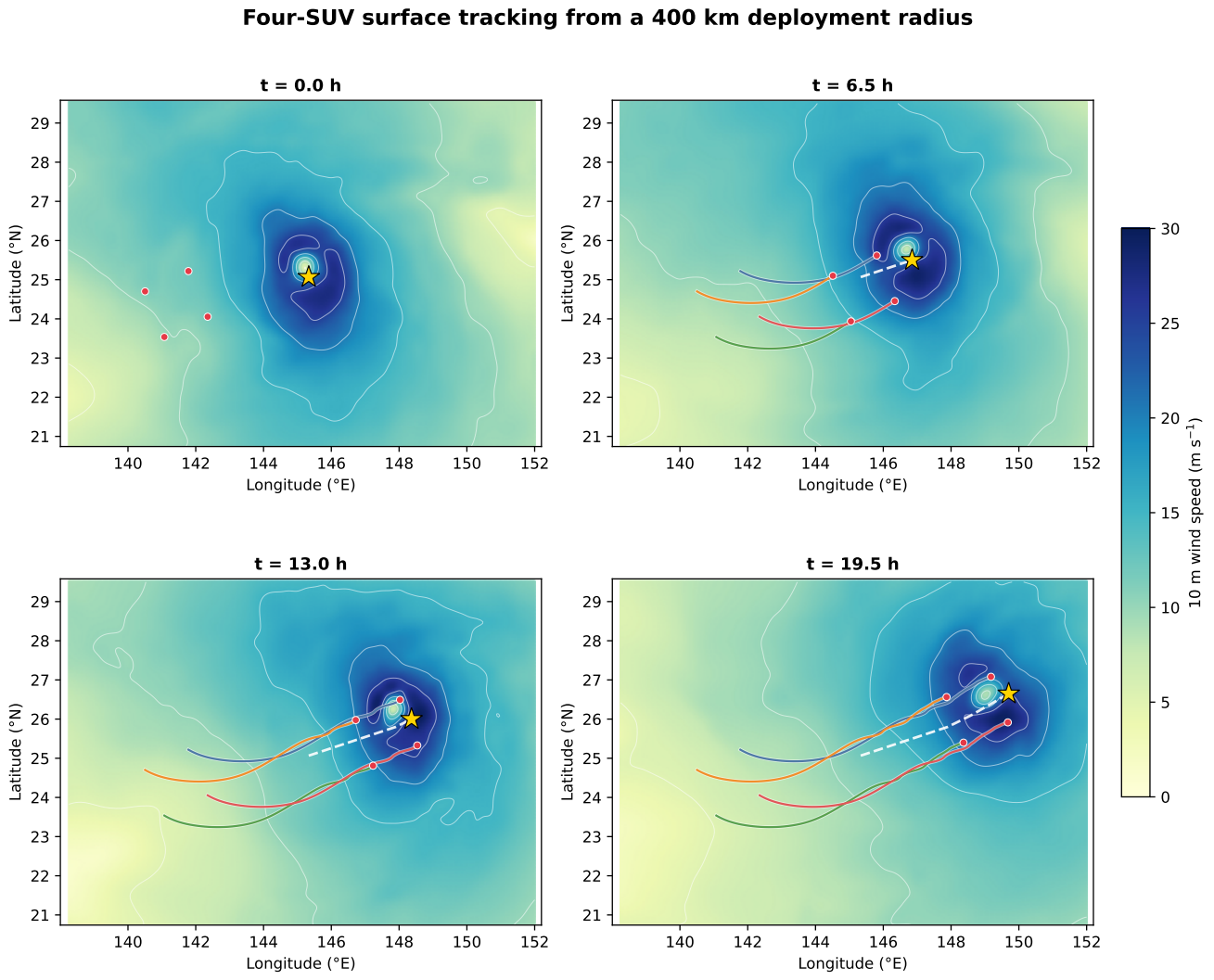


Figure 2: Four-SUV surface-tracking process for CHAMPI (2015) at the 20 m/s speed limit. The four panels show the initial deployment and subsequent states at approximately 6.5, 13.0, and 19.5 h. The color background is the spatially interpolated ERA5 10 m wind-speed field at each time. Small red dots mark vehicle positions, and the yellow star marks the TC center.

The relative time-averaged distance penalty of the 12 m/s policy is +6.3% at 100 km, +10.6% at 200 km, +18.2% at 400 km, +17.7% at 600 km, and +14.0% at 800 km. The non-monotonic relative pattern, together with the monotonic increase in absolute final-tracking-error benefit, is consistent with the finite time available to remove forecast-induced displacement. Overall, the results show that the policy learned from wind observations alone provides increasing operational value for large initial displacement errors when a 20 m/s speed envelope is available, while the 12 m/s envelope offers a lower-speed, shorter-path alternative.

Taken together, Figs. 3–4 and Table 5 connect the operational benefit metric to both population-level statistics and representative trajectories. The right panel of Fig. 3 is the primary benefit comparison: for each deployment radius, the preferred configuration is the one with the smallest final tracking error. Figure 2 shows the mechanism in the four-SUV, 400 km case, where the formation first turns along the observed wind structure and then reduces its separation from the TC center. Figure 4 shows that this inward correction persists from 100 to 800 km, although the final marker remains farther from the center as the initial deployment error increases. These representative paths explain why an effective controller can reduce error without making the final tracking error independent of deployment radius.

Table 5 provides a same-policy check of the benefit interpretation. Applying the execution filter lowers the fleet-mean final tracking error from 346.4 to 292.6 km, a 53.8 km improvement, while also shortening trajectories and reducing mean and peak speed. Thus the lower final tracking error, and therefore the higher operational observation benefit, is not obtained by simply commanding more aggressive propulsion. The trajectory figures and the ablation table jointly indicate that better final placement results from smoother closed-loop correction of the observed wind pattern.

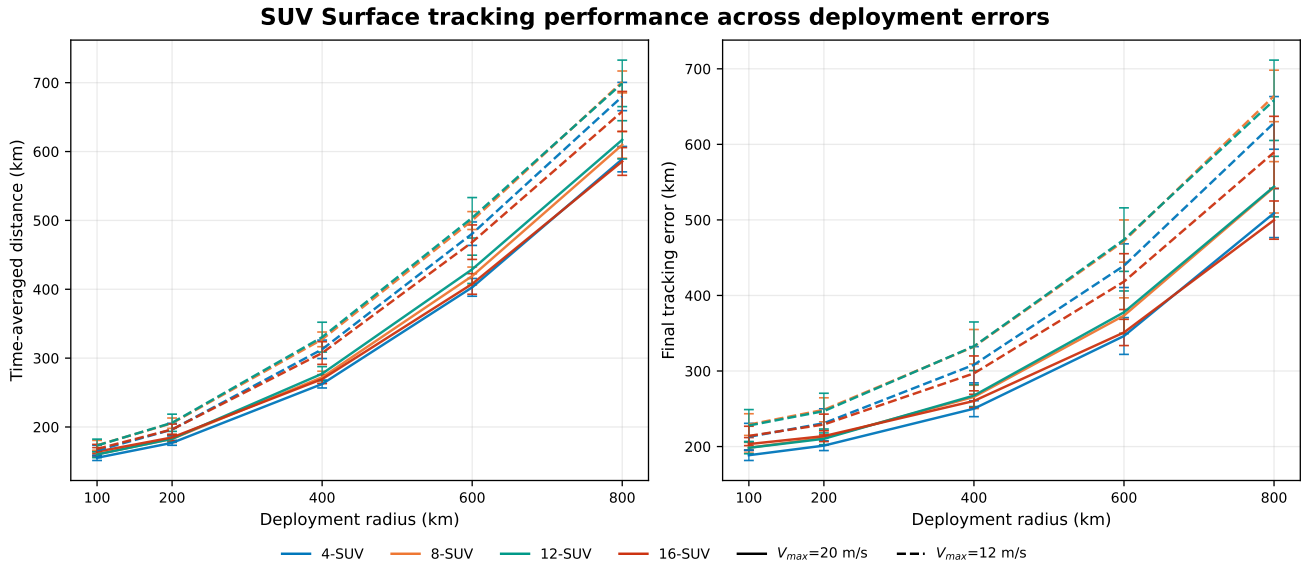


Figure 3: SUV surface tracking performance across deployment errors. The left panel shows the fleet-mean time-averaged distance to the TC center, representing whether the fleet remains close to the moving TC throughout an episode. The right panel shows the final tracking error to the TC center at the end of the episode; smaller values indicate higher operational observation benefit. Colors denote fleet sizes and line styles denote the 20 and 12 m/s speed limits. Each point summarizes ten independent training runs (ten random seeds), with error bars showing the sample standard deviation.

To provide an internally consistent visual comparison, Figs. 2–6 use the same held-out CHAMPI (2015) segment. The surface figures show motion from the segment start to its endpoint, while the subsurface rollout starts from the segment-end anchor, representing the sequential transition from surface delivery to underwater observation.

Figure 2 resolves the temporal evolution for the four-SUV, 400 km case. At deployment, the four vehicles are dispersed around the formation center far from the TC. During the first several hours, the vehicles turn toward the strengthening wind region and begin to reduce the initial displacement. By 6.5 h, all four vehicles have initiated the inward turn; by 13.0 h they have entered the outer high-wind region; and by 19.5 h all four trajectories have turned into the high-wind branch and substantially reduced their initial displacement. The individual paths differ because each vehicle has its own state and position within the shared fleet-level wind observation; nevertheless, the fleet-level motion follows the wind pattern toward the TC core and illustrates long-range error correction in this representative episode.

Figure 4 shows how initial forecast error changes the surface search process. At 100–200 km, the formations converge rapidly and remain concentrated around the TC core. As the deployment radius increases to 600–800 km, the trajectories lengthen and fan out over a wider region before turning toward the TC. Increasing the fleet from 4 to 16 SUVs produces a denser set of sampled branches and broader spatial coverage, but it does not remove the need for long-range transit. In the 12- and 16-SUV cases, many paths overlap near the core because the shared wind observations guide the vehicles toward the same high-wind region. In these representative cases, fleet size mainly broadens the sampled approach region, whereas deployment error primarily determines the required transit distance.

The 600 and 800 km rows also reveal two coupled failure mechanisms at large initial displacement. At 600 km, the representative fleet can still recover a usable inward direction from the observed wind-speed field and its spatial gradient, so the vehicles continue toward the TC despite the longer transit. At 800 km, the vehicles must first overcome a larger geometric separation with finite speed and a finite episode duration; in addition, the TC-induced wind speed and its local gradient become weaker and less distinctive at that range. The reduced signal quality makes the approximate bearing toward the TC more ambiguous, so the policy may approach more slowly, change direction repeatedly, or fail to recover the TC within the episode. These two effects are coupled: longer travel time increases the consequence of even a small directional error, while the weaker remote wind signature makes such an error more likely.

5.2. Surface Formation Comparison

The ring, line, and grid formations provide three different ways to allocate spatial observations across eight SUVs. Each was trained with the same 2-million-step protocol, deployment distribution, and safety layer at both 12 and 20 m/s, then evaluated without retraining at 100–800 km (Table 4). Averaged over the ten speed–radius conditions, the ring gives the lowest mean final error (346.7 km), followed by the grid (356.6 km) and the line (364.4 km). The row-wise result is condition-dependent: at 12 m/s, the line is best at 100 km and the ring is best at 200–800 km; at 20 m/s, the line is best at 100–400 km, the grid at 600 km, and the ring at 800 km.

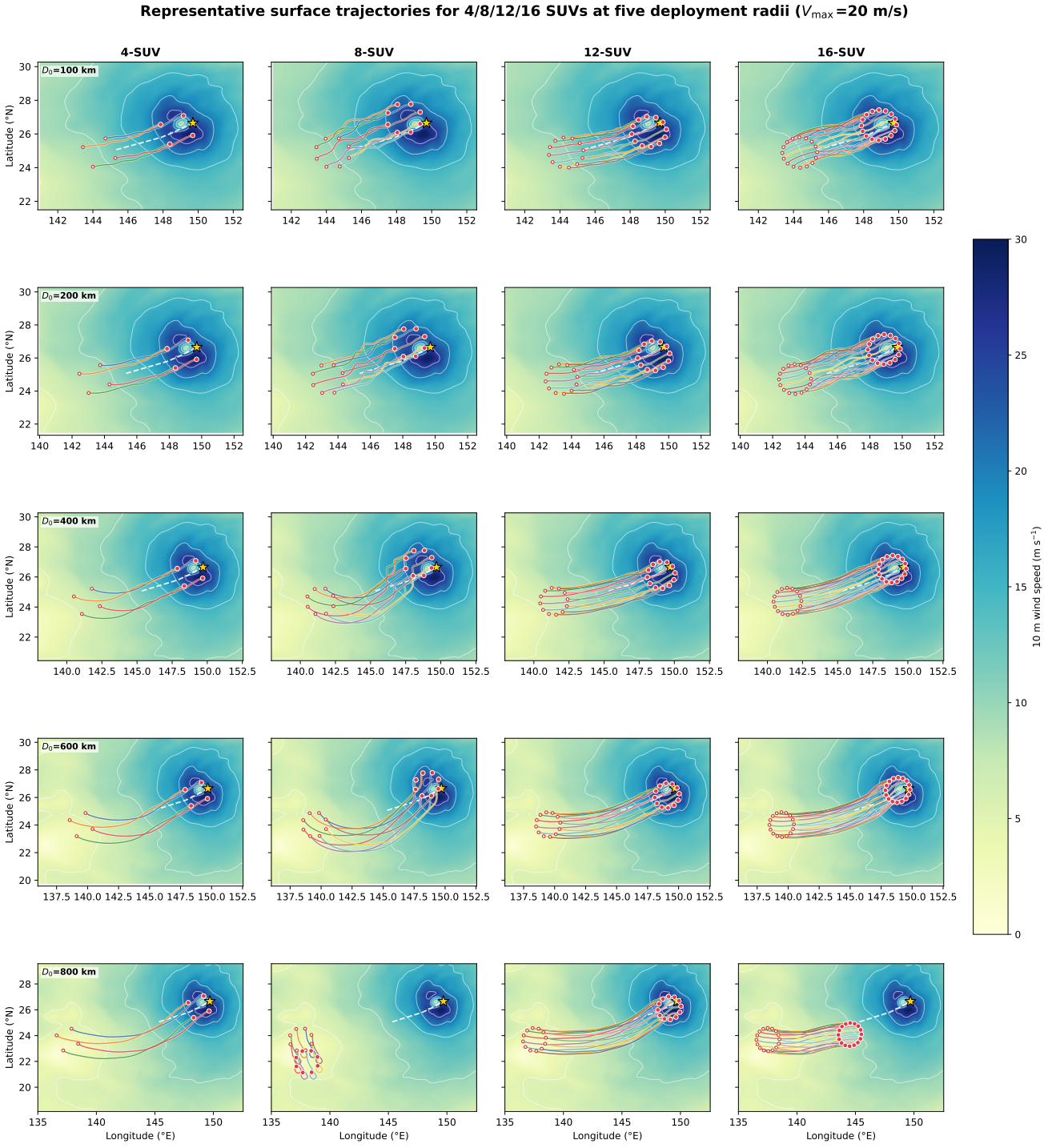


Figure 4: Representative CHAMPI (2015) trajectories for the final 20 m/s surface policy. Columns show 4-, 8-, 12-, and 16-SUV formations, while rows show nominal formation-center deployment radii of 100, 200, 400, 600, and 800 km. The color background is the spatially interpolated final-time ERA5 10 m wind-speed field. Thin colored lines are vehicle trajectories, open circles mark initial positions, small red dots mark final vehicle positions, and the yellow star marks the TC center at the end of the episode.

The line formation produces the shortest per-vehicle path in all ten conditions (470.3 km on average versus 605.3 km for the ring), but it contracts the mean formation radius to 57.1 km and the pairwise spacing to 85.2 km. The ring retains a 99.9 km radius and 143.6 km spacing, giving a broader observation footprint. The grid preserves a comparable footprint (103.6 km radius and 156.8 km spacing) but improves the row-wise final error only for the 20 m/s, 600 km case and has longer paths than the line in every condition. We therefore retain the ring for the main broad-coverage configuration, while treating the line as a transit-efficient

Table 4
Surface initial-formation comparison for eight SUVs. Each policy was trained with formation-center offsets sampled inside an 800 km radius and evaluated without retraining at fixed offsets of 100, 200, 400, 600, and 800 km on the same 768 held-out TC segments. Error is the final SUV-to-best-track-center distance and path is the per-vehicle trajectory length. Bold entries are the best value in each speed/radius row; error and path are assessed separately. All distances are in km.

Speed	Train radius	Eval radius	Ring		Line		Grid	
(m/s)	(km)	(km)	Error	Path	Error	Path	Error	Path
12	800	100	225.8	398.1	**222.9**	**325.9**	236.7	367.0
12	800	200	**250.6**	428.4	253.9	**335.7**	257.0	398.3
12	800	400	**335.9**	475.0	368.1	**321.8**	345.5	420.6
12	800	600	**480.7**	499.0	523.9	**301.6**	487.8	423.5
12	800	800	**648.1**	505.6	733.2	**273.6**	691.7	410.7
20	800	100	197.2	599.6	**164.5**	**538.5**	203.0	615.0
20	800	200	209.8	653.7	**179.6**	**583.7**	213.8	656.2
20	800	400	262.7	765.2	**250.9**	**651.8**	258.1	743.4
20	800	600	354.4	840.2	381.4	**677.5**	**353.3**	792.9
20	800	800	**501.9**	888.0	566.1	**693.2**	518.7	830.8

alternative and the grid as a spacing-preserving sensitivity case rather than a universally superior formation.

Figure 5 visualizes one matched CHAMPI segment. The formation geometry is not rigidly preserved because each policy moves under the same shared wind observation but with its own state. This representative episode illustrates how a compact line can move efficiently, whereas the ring and grid retain a broader sampling footprint. The aggregate 768-segment results, rather than the selected episode, determine the formation choice. We therefore retain the ring as the main configuration and report the alternatives as a sensitivity study rather than as evidence that one formation is universally optimal.

5.3. Execution-Layer Action Filtering

Table 5 reports the low-pass filtering ablation over the same trained actor, validation dataset, and 10 min decision interval. Without filtering, the 4-, 8-, 12-, and 16-SUV configurations travel 1294.9–1331.5 km per vehicle, with mean speeds of 19.23–19.73 m/s and maximum speeds of 19.92–20.00 m/s. With the execution-layer filter ($\beta = 0.8$), path length falls to 690.8–770.9 km, mean speed to 11.11–12.72 m/s, and maximum speed to 16.13–18.20 m/s. The final tracking error changes from 288.2 to 258.2 km for four SUVs, from 305.2 to 308.6 km for eight SUVs, from 478.5 to 330.0 km for 12 SUVs, and from 313.6 to 273.4 km for 16 SUVs. Averaged over the four fleet sizes, it decreases from 346.4 to 292.6 km, a reduction of 53.8 km (15.5%). Thus, filtering not only avoids degrading final tracking but improves it on average: three fleet sizes improve, whereas the eight-SUV case increases only slightly, by 3.4 km.

The same comparison shows reductions for every fleet size: path length decreases by 563.4, 547.1, 618.5, and 573.6 km for 4, 8, 12, and 16 SUVs, respectively, while maximum speed decreases by 1.80, 2.29, 3.87, and 2.94 m/s. The 12-SUV configuration has the largest relative path-length reduction, although the effect is not monotonic in fleet size. These results show that the execution filter improves final tracking on average while substantially shortening trajectories and reducing mean and peak speed.

5.4. Subsurface Strategy Comparison

For each depth, four SUVs seek higher current speed; the central comparison is therefore the fleet-mean current speed experienced along each trajectory, together with its path and propulsion cost. The comparison follows the protocol in Section 4.5. Figure 6 first illustrates the resulting trajectory structure for the three strategies.

Figure 6 illustrates the qualitative trajectory construction for the three strategies at shallow, intermediate, and deep sampling layers. Across these three representative depths, the mean experienced current speeds of active control, random control, and passive drift remain close within each depth, whereas their path lengths differ substantially: active control requires the longest paths and passive drift the shortest. The background fields also show that the spatial current structure changes with depth, which is why a single-depth example is insufficient. These episodes remain illustrative; all comparative inference is deferred to the final ten-seed, seven-depth evaluation.

Figure 7 reports the final ten-seed results at all seven depths from 0 to 60 m. At every depth and both propulsion limits, active IPPO attains a higher mean experienced current speed than passive drift and random control. The improvement over passive drift ranges from 2.1% to 5.4%, while the corresponding improvement over random control ranges from 2.0% to 5.1%. Random control remains close to passive drift in mean current speed but travels substantially farther.

The longer trajectories of active IPPO are an intended observation advantage, not merely a motion cost. Its path is 3.6–7.3 times that of passive drift, allowing the four SUVs to move through a larger spatial footprint and acquire more current profiles and contextual samples. The corresponding nominal coverage increases by roughly 3.3–5.7 times. Across the 140 seed-configuration comparisons, the active

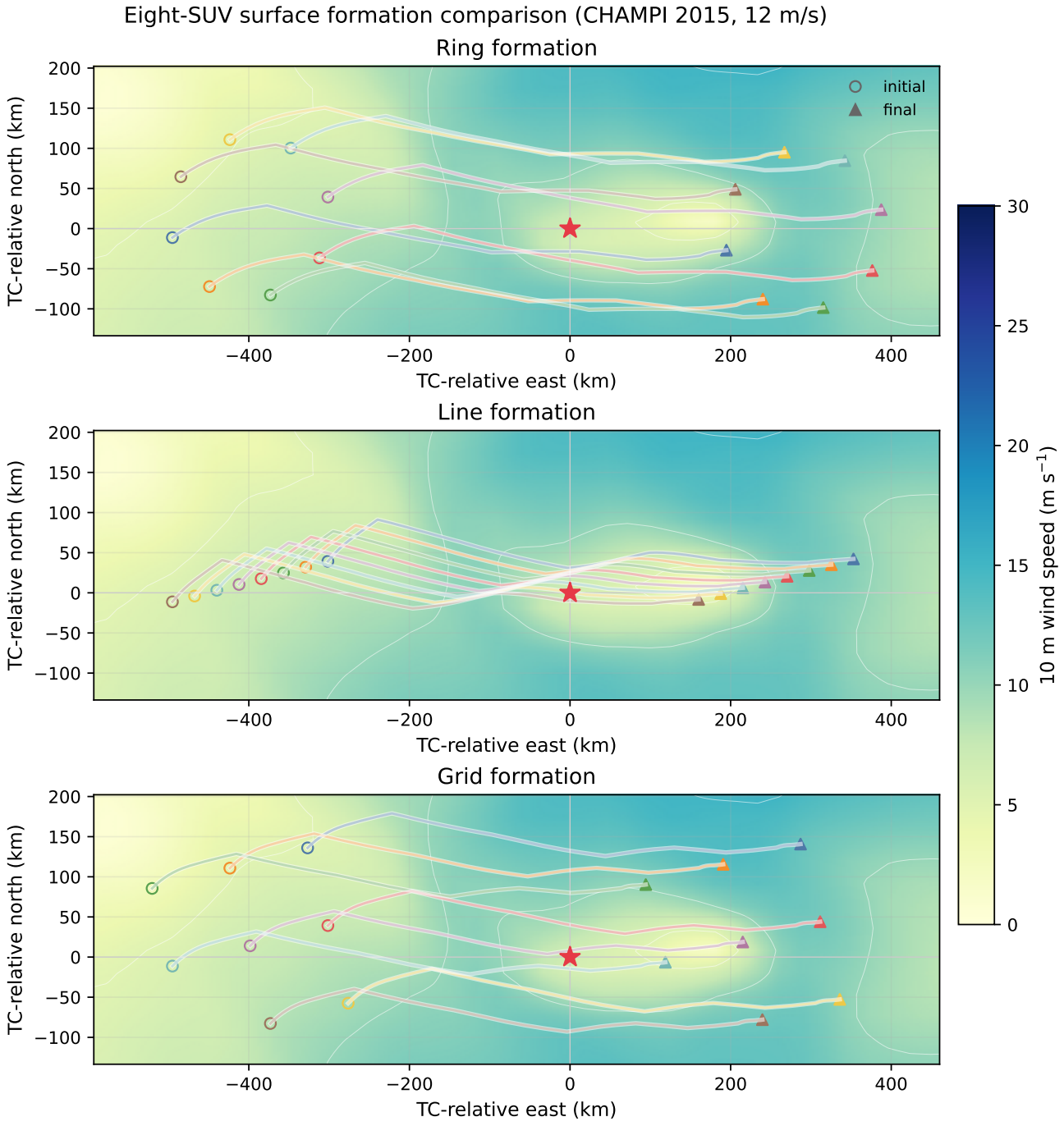


Figure 5: Representative surface trajectory comparison for ring, line, and grid formations with eight SUVs at 12 m/s. Open circles denote initial positions, triangles denote final positions, and the red star denotes the TC center. The color background and contours show the spatially interpolated ERA5 10 m wind-speed field at the final simulation step. Axes are relative to the instantaneous TC center.

policy has a higher mean experienced current speed in 133 cases; uncorrected paired tests classify 117 as significant improvements and three as significant deficits. Therefore, active control is not judged only by a small increase in mean current speed: it also increases the amount and spatial extent of the collected underwater data. Passive drift remains the low-motion reference and cannot deliberately relocate toward higher-current regions.

The propulsion limit consistently affects trajectory extent. At all seven depths, the 2.0 m/s policy travels 1.24–1.31 times farther than the 1.5 m/s policy because the larger speed envelope permits more movement within the same 20 h episode. This result links future platform speed directly to

Table 5
Execution-layer low-pass filtering for 4/8/12/16 surface SUVs on 768 held-out segments from 50 unseen TCs in the 500-TC dataset. Each trained actor is evaluated with the filter off and on ($\beta = 0.8$) on the same validation episodes.

Fleet	Filter	Final tracking error (km)	Path per SUV (km)	Mean speed (m/s)	Maximum speed (m/s)
4	Off	288.2	1331.5	19.73	20.00
4	On	**258.2**	**768.1**	**12.72**	**18.20**
8	Off	**305.2**	1318.0	19.57	19.97
8	On	308.6	**770.9**	**12.58**	**17.68**
12	Off	478.5	1309.3	19.34	20.00
12	On	**330.0**	**690.8**	**11.11**	**16.13**
16	Off	313.6	1295.0	19.23	19.92
16	On	**273.4**	**721.4**	**11.94**	**16.98**

Final tracking error is the fleet average of the individual SUV-to-best-track-center distances at the final time step. Off/on changes only the execution-layer filter on the same trained actor.

observation capacity: a faster SUV can cover more of the TC-induced current field and collect more profiles, although the associated energy and control constraints must still be considered. Mean experienced drift current decreases from 0.521 to 0.292 m/s between 0 and 60 m, and passive-drift paths contract from 33.6 to 17.1 km. Active paths decrease more moderately, from 122.1 to 98.7 km at 1.5 m/s and from 155.1 to 124.6 km at 2.0 m/s. The increasing active/passive path ratio with depth therefore reflects both weaker ambient advection and the largely depth-independent propulsion envelope, rather than an increasing policy advantage.

The current fields in Fig. 6 contain pronounced horizontal gradients and localized high-speed bands. The fleet-mean experienced current speed in the final experiments ranges from 0.29 to 0.54 m/s across methods and depths, but this average includes the full sampled trajectory rather than only the strongest-current patches. Local current speeds can therefore be substantially higher than the fleet mean and are of comparable magnitude to the 1.5 and 2.0 m/s propulsion limits. Active propulsion can relocate a vehicle toward such patches, whereas passive drift follows the ambient flow without steering. The completed comparisons show that active control uses this capability to obtain both a modestly higher mean current speed and a substantially larger observation footprint and data volume.

5.5. Comparison with Traditional Planners

Table 6 summarizes the comparison with four traditional planners. Across all 40 matched surface configurations, parameter-sharing IPPO attains the lowest final tracking error and time-averaged distance. Relative to the potential-field, A*, lawnmower, and information-greedy planners, its mean final error is reduced by 74.8, 117.3, 213.1, and 125.3 km, respectively. The IPPO path is shorter than those of the first two baselines in all 40 comparisons. It is also shorter than the lawnmower and information-greedy paths in 28 and 29 of the 40 comparisons; their fleet-average paths are still longer, but the path advantage is less uniform. This result indicates that the learned controller preserves direct inward motion while adapting more effectively to local wind structure than the local-gradient and rolling-map planners.

Underwater, the four baselines are not dominated on every metric. The potential-field planner attains the highest mean experienced current (0.449 m/s), followed by information-greedy control (0.432 m/s) and A* (0.428 m/s); IPPO and lawnmower coverage are close to each other at approximately 0.409 m/s. A* instead has the shortest mean path (94.8 km), whereas IPPO produces the largest nominal coverage (3630 km^2, compared with 1976, 2275, 2902, and 1520 km^2 for potential field, A*, lawnmower, and information-greedy control, respectively). The episode-level current-weighted coverage $C_w(z)$ is nevertheless larger for IPPO than for every baseline in all 14 depth–speed settings after paired testing, by factors of 1.52–1.71 for potential field, 1.40–1.59 for A*, 1.24–1.34 for lawnmower coverage, and 2.07–2.54 for information-greedy control. IPPO therefore provides the broadest current-weighted observation, but not the strongest value in every individual underwater metric.

Appendix Table A.1 resolves these averages by deployment radius and ocean depth. IPPO retains the lowest final error and time-averaged distance at all five surface radii. Underwater, the metric-specific ordering is consistent at every depth: potential field gives the highest mean experienced current, IPPO gives the largest nominal coverage, and A* gives the shortest path. The aggregate rankings are therefore not driven by a single radius or depth.

6. Conclusion

This study introduced a simulation-based parameter-sharing IPPO framework for coordinated surface–subsurface tropical-cyclone observation with Blue Whale-class SUVs. The 12 m/s surface and 1.5 m/s submerged limits approximate the current Blue Whale design speeds, whereas 20 m/s and 2.0 m/s represent possible future upgrades; these near-current and future-upgrade settings were compared within their respective tasks. The 12 and 20 m/s surface envelopes cover 93.8% and 98.9% of the sampled IBTrACS TC translation speeds, respectively, whereas the submerged envelopes are intentionally used for local sampling rather than TC chasing. Event-specific environments aligned

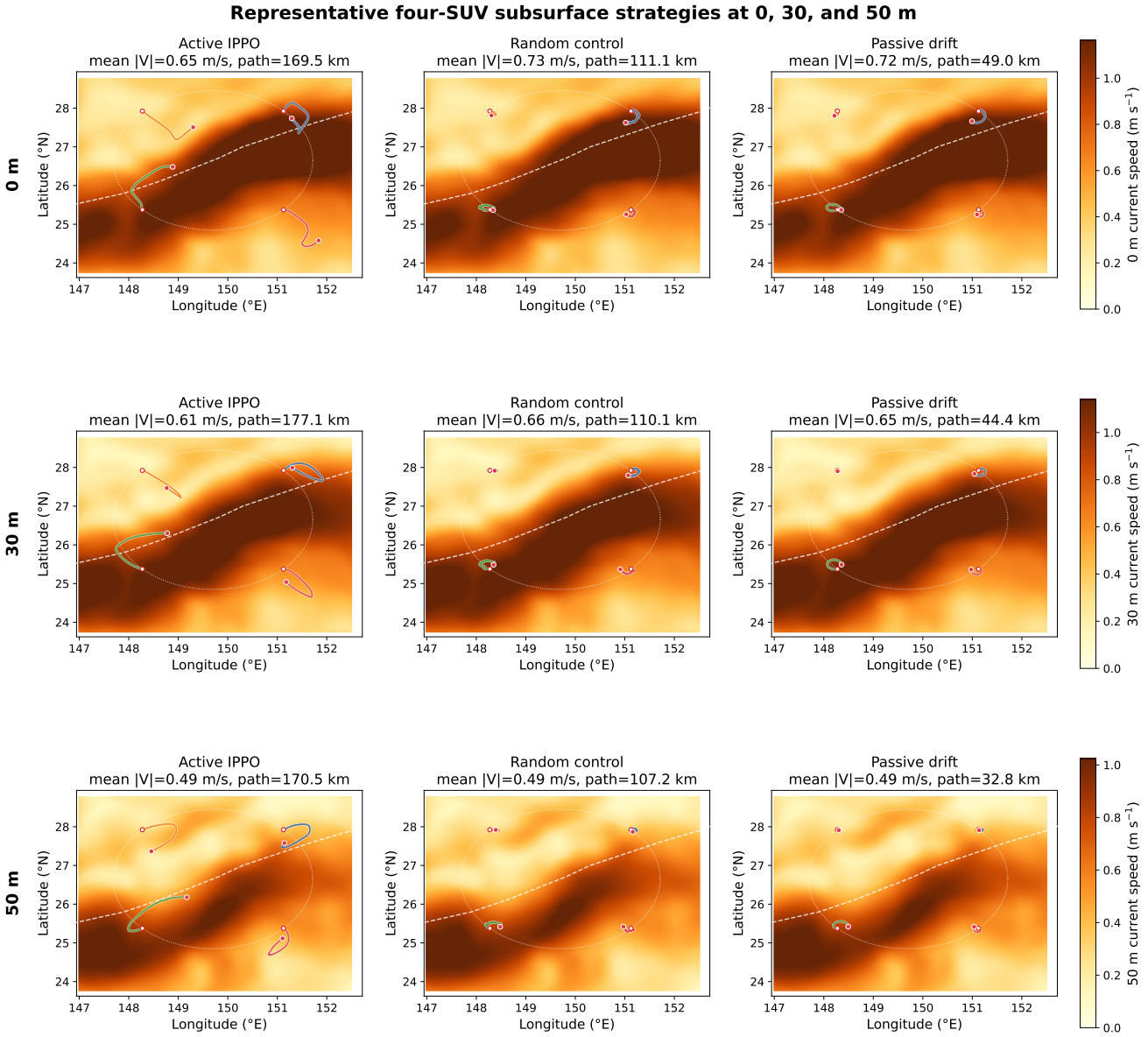


Figure 6: Representative four-SUV trajectories at 0, 30, and 50 m for the same CHAMPI (2015) segment used in the surface trajectory figures. These three depths provide shallow, intermediate, and deep examples from the full 0–60 m evaluation matrix. Rows denote depth and columns denote Active IPPO, random control, and passive drift. The propulsion limit is 2.0 m/s and the episode duration is 20 h. Background color is the spatially interpolated episode-mean HYCOM current speed at the corresponding depth; dashed white curves show the TC best track, open circles mark the initial vehicle positions, the dotted circle marks the 200 km initialization boundary, and small red arrows mark final vehicle positions.

ERA5 10 m winds, HYCOM currents at seven depths from 0 to 60 m, and IBTrACS tracks for 500 TCs, with 450 TCs used for training and 50 unseen TCs for testing.

The surface and subsurface tasks have different objectives. At the surface, the fleet uses shared wind observations to follow the TC and reach its high-wind region; the final tracking error measures how close the randomly deployed SUVs finish to the TC center. Across ten seeds, the 20 m/s policy achieved smaller time-averaged tracking distances than 12 m/s, and its advantage increased with deployment error. The 12 m/s policy instead produced shorter paths and lower speeds. Low-pass filtering reduced the fleet-mean final tracking error from 346.4 to 292.6 km while also shortening trajectories and reducing peak speed. These results show that the closed observation–decision–motion loop can guide the fleet inward using wind observations alone during validation.

At depth, four SUVs are assigned to each layer and seek the strongest observable current rather than the surface-center projection. Active IPPO, random control, and passive drift are compared under identical episodes and propulsion limits. Across all seven depths, active IPPO raises the mean experienced current speed by 2.1–5.4% and produces trajectories 3.6–7.3 times longer than passive drift, together

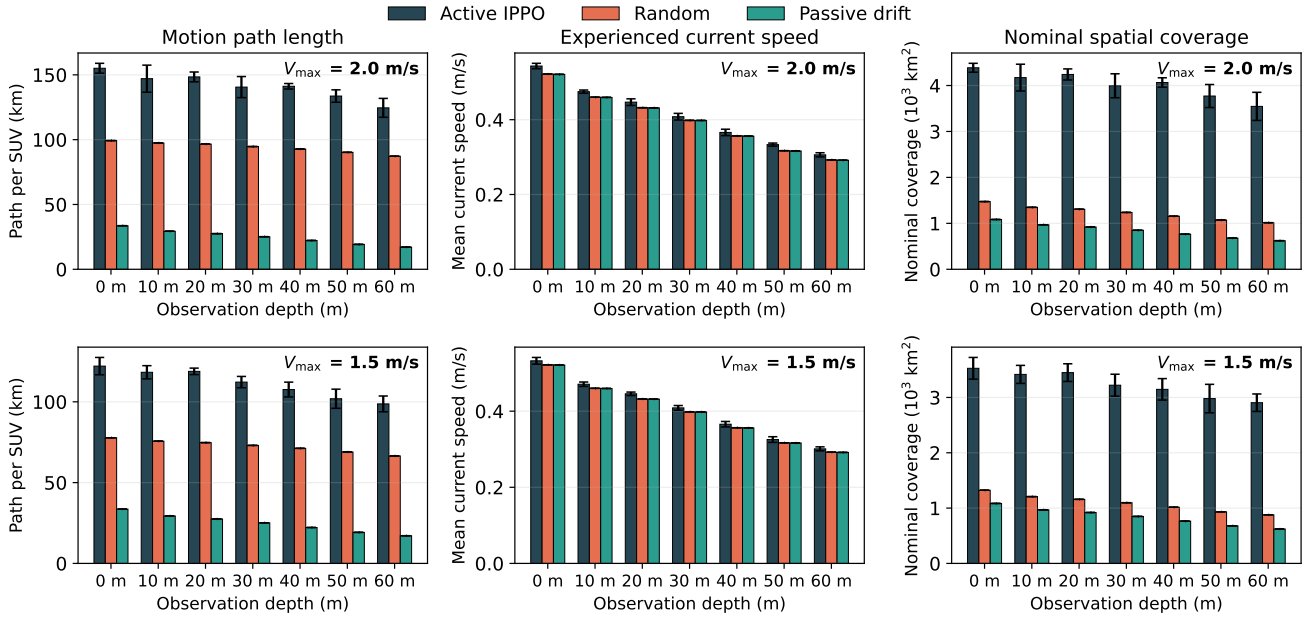


Figure 7: Comparison of active control, random control, and passive drift in the 20 h subsurface task at seven depths from 0 to 60 m. Rows show maximum propulsion speeds of 2.0 and 1.5 m/s; columns report the fleet-mean motion path length, mean experienced current speed, and nominal spatial coverage area. Error bars show the sample standard deviation across ten active-policy training seeds. Random control and passive drift are paired baselines without training-seed variability.

Table 6
Comparison with four traditional planning baselines on the held-out TC segments from the same 500-TC train/test split. Surface values are means over the 4/8/12/16-SUV fleets, both speed limits, and all five deployment radii. Subsurface values are means over all seven depths and both propulsion limits. Smaller errors and paths are better; higher current and coverage are better. No single method is best in all subsurface columns.

Panel A: surface tracking			
Method	Final error (km)	Time-averaged distance (km)	Path per SUV (km)
Parameter-sharing IPPO	**341.7**	**349.1**	**463.6**
Potential field	416.5	405.9	727.6
Receding-horizon A*	459.0	430.4	824.9
Lawnmower coverage	554.8	478.4	508.4
Information greedy	467.0	429.6	525.2
Panel B: subsurface observation			
Method	Mean current (m/s)	Nominal coverage (km^2)	Path (km)
Parameter-sharing IPPO	0.409	**3630**	126.4
Potential field	**0.449**	1976	112.5
Receding-horizon A*	0.428	2275	**94.8**
Lawnmower coverage	0.409	2902	111.1
Information greedy	0.432	1520	102.4

with about 3.3–5.7 times greater nominal spatial coverage. The learned propulsion therefore serves two purposes: it guides the SUVs toward higher-current water and keeps them moving over a wider area, producing more profiles and more contextual observations than a drift-only strategy. Compared with the four traditional planners, IPPO attains the lowest final tracking error in all 40 matched surface configurations. Underwater, potential field attains the highest mean local current and A* the shortest path, whereas IPPO attains the largest current-weighted spatial coverage in all 14 settings. Its advantage therefore comes from the combination of current exposure and broad spatial sampling rather than metric-wise dominance.

The 2.0 m/s policy travels 1.24–1.31 times farther than the 1.5 m/s policy at every depth. A larger speed envelope permits more movement within the same 20 h evaluation

horizon and therefore creates more opportunities to observe TC-induced currents. This identifies a direct path for future platform development: increasing SUV speed could raise the observation benefit for subsurface temperature, salinity, current, and biogeochemical fields, provided that energy use and control stability remain acceptable. The full seven-depth matrix therefore supports active sampling as an observation-capacity strategy while also exposing the propulsion-cost trade-off that must be managed operationally.

The present evidence is simulation-based and uses a mission-level horizontal point-mass model. Field validation, operational forecast impact, and calibrated six-degree-of-freedom vehicle dynamics remain future work. The results show generalization to unseen TCs within the simulated western North Pacific environment, but they do not constitute a guarantee of sim-to-real performance.

Code and package availability

The code and processed results are available at `https://github.com/DYHAI/SUV-MARL-TC`. The experiments used Python 3.11, PyTorch 2.11.0 (CUDA 12.8 build), Stable-Baselines3 2.9.0, and Gymnasium 1.2.2 for policy training; NumPy 2.4.6 and pandas 2.3.3 for numerical and tabular processing; SciPy 1.17.1, xarray 2026.7.0, netCDF4 1.7.4, and h5py 3.16.0 for analysis and NetCDF processing; and Matplotlib 3.11.1 for visualization. All policy training was performed on CPU.

Acknowledgements

This project is supported by the National Natural Science Foundation of China (42530402 and 42227901), the National Natural Science Foundation of China Youth Science Fund Project (62603673), the China Postdoctoral Science Foundation (2025M783467), the Project of State Key Laboratory of Satellite Ocean Environment Dynamics (SOEDZZ2527), and the Innovation Group Project of Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) (311024004).

A. Reward and IPPO Hyperparameters

Table A.1 reports the traditional-planning comparison with the two surface distance metrics separated by deployment radius and the three subsurface observation metrics separated by depth.

Table A.1: Distance and observation metrics for the traditional planner comparison on the held-out TC segments. Panel A reports the two surface distance metrics by deployment radius, averaged over the four fleet sizes and both speed limits. Panel B reports mean experienced current speed, nominal coverage, and path per SUV at each depth, averaged over the two propulsion limits.

Panel A: surface distance metrics						
Radius	Metric	IPPO	PF	A*	Lawn	Greedy
100 km	Final error (km)	**209.0**	290.9	325.9	366.0	296.0
	Time-averaged distance (km)	**165.0**	230.0	250.6	232.9	212.4
200 km	Final error (km)	**224.1**	294.9	329.9	394.5	311.0
	Time-averaged distance (km)	**191.5**	248.2	267.1	285.9	244.3
400 km	Final error (km)	**289.2**	354.1	395.2	504.5	406.2
	Time-averaged distance (km)	**294.9**	343.1	367.7	441.7	374.3
600 km	Final error (km)	**406.5**	477.5	531.5	661.2	560.4
	Time-averaged distance (km)	**451.4**	506.7	538.8	621.7	559.0
800 km	Final error (km)	**579.5**	665.1	712.6	847.6	761.5
	Time-averaged distance (km)	**642.4**	701.5	727.8	809.7	757.8
Panel B: subsurface metrics by depth						
Depth	Metric	IPPO	PF	A*	Lawn	Greedy
0 m	Mean experienced current (m/s)	0.538	**0.575**	0.551	0.538	0.553
	Nominal coverage (km^2)	**3957**	2189	2421	3047	1556
	Path per SUV (km)	138.6	119.8	**101.3**	116.0	104.8
10 m	Mean experienced current (m/s)	0.473	**0.513**	0.490	0.474	0.495
	Nominal coverage (km^2)	**3794**	2137	2387	3004	1590
	Path per SUV (km)	132.7	118.1	**99.9**	114.8	105.5
20 m	Mean experienced current (m/s)	0.446	**0.484**	0.463	0.446	0.467
	Nominal coverage (km^2)	**3845**	2094	2351	2979	1595
	Path per SUV (km)	133.6	116.5	**98.1**	113.9	104.9
30 m	Mean experienced current (m/s)	0.408	**0.451**	0.430	0.411	0.435
	Nominal coverage (km^2)	**3608**	2017	2288	2918	1558
	Path per SUV (km)	126.4	113.3	**95.0**	111.8	103.1
40 m	Mean experienced current (m/s)	0.366	**0.409**	0.387	0.369	0.394
	Nominal coverage (km^2)	**3606**	1923	2241	2864	1517
	Path per SUV (km)	124.4	110.3	**92.7**	109.9	101.5
50 m	Mean experienced current (m/s)	0.329	**0.369**	0.351	0.328	0.354
	Nominal coverage (km^2)	**3375**	1786	2157	2805	1445
	Path per SUV (km)	117.8	107.0	**89.9**	107.7	99.7
60 m	Mean experienced current (m/s)	0.304	**0.340**	0.325	0.301	0.326
	Nominal coverage (km^2)	**3225**	1683	2083	2693	1381
	Path per SUV (km)	111.6	103.0	**86.8**	103.7	97.3

Table A.2 summarizes the reward, action-filter, and safety parameters used in the surface and underwater experiments. Table A.3 lists the shared IPPO network and training hyperparameters.

Table A.2: Surface and subsurface reward, action-filter, and coastal-safety parameters used in the final experiments.

Parameter	Symbol	Value	Description and applicable task
Panel A: surface reward and controller parameters			
Proximity-cost distance cap	$D_{\max}$	300 km	Distance cost is capped at 300 km to prevent excessively large gradients from distant initial deployments.
Distance normalization scale	D_{scale}	100 km	Normalization scale used by the proximity and radius-excess reward terms.
Near-center threshold	d_{near}	100 km	Threshold defining the discrete near-center bonus.
Near-center bonus	b_{near}	1.0	Bonus assigned when an SUV is within 100 km of the TC center.
Radius-excess threshold	R_{form}	100 km	Reference radius used by the additional distance penalty beyond 100 km.
Radius-excess weight	W_{form}	1.0	Weight of the penalty applied when an individual SUV is farther than 100 km from the TC center.
Action-change weight	λ_u	0.05	Weight of the squared action change $\|\bar{\mathbf{a}}_{i,k} - \bar{\mathbf{a}}_{i,k-1}\|_2^2$ between successive executed commands.
Action-filter coefficient	β	0.8	Coefficient of $\mathbf{a}_k^{\text{filt}} = (1-\beta)\mathbf{a}_k^{\text{raw}} + \beta\mathbf{a}_{k-1}^{\text{filt}}$. This is a controller setting rather than a reward term.
Surface reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 2]$	Clipping range applied to the team-averaged surface reward.
Panel B: underwater active-control comparison parameters			
Current-speed normalization	V_{norm}	1.0 m/s	Current-intensity reward is implemented as $\min(\|\mathbf{u}\|/V_{\text{norm}}, R_{\max})$.
Current-reward cap	$R_{\max}$	2.0	Upper bound of the normalized current-speed reward.
Downstream-alignment weight	$\beta_{\parallel}$	0.5	Weight of the positive propulsion–current alignment term.
Underwater reward clipping range	$[r_{\min}, r_{\max}]$	$[-5, 5]$	Clipping range applied to the team-averaged underwater reward during active control training.
Panel C: coastal-safety parameters by stage			
Land penalty	P_{land}	20	Penalty applied when an SUV enters a land cell in either stage.
Nearshore buffer distance	D_{buf}	100 km	Width of the soft nearshore penalty buffer in both stages.
Nearshore penalty weight	W_b	8.0	Maximum soft-penalty weight within the 100 km nearshore buffer in either stage.
Minimum deployment offshore distance	D_{deploy}	100 km	Minimum coastline distance required by offshore-only rejection sampling during surface deployment; it is not applied to subsurface initialization.
Action-filter safety margin	D_{filter}	100 km	Surface-stage distance below which the onshore component of the commanded action is removed. The subsurface comparison does not use this filter.

Note: the final subsurface comparison consists of active IPPO, random control, and passive drift. Passive drift does not optimize a learned propulsion policy or reward.

Table A.3: Parameter-sharing IPPO network, training, and deployment hyperparameters used in the final surface and subsurface experiments.

Parameter	Symbol	Value	Description
Algorithm	–	PPO	Stable-Baselines3 PPO with parameter sharing and independent per-agent action sampling.
Return aggregation	–	Team average	The environment reward is averaged across the N SUVs before computing the shared PPO return.
Optimizer	–	Adam	Adam optimizer used for the actor and critic parameters.
Policy network	–	2×256 MLP	Actor: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh–Linear(256, d_{act}).
Value network	–	2×256 MLP	Critic: Linear(d_{obs}, 256)–Tanh– Linear(256, 256)–Tanh–Linear(256, 1). Per-agent values are summed during training.
Per-agent observation dimension	d_{obs}	14/26/38/50	Surface wind observations for 4/8/12/16 SUVs, respectively. The four-SUV underwater active-control comparison uses $d_{\text{obs}} = 17$.
Surface observation	–	Wind at 10 m above sea level	Using ideal surface communication, each SUV receives the current wind speed and components at 10 m above the sea surface observed by every SUV in the fleet.
Underwater observation	–	Current only	With no inter-vehicle communication, the active IPPO controller observes only its own local current speed and components at the assigned ocean depth, together with a five-frame history and its own normalized propulsion velocity.
Action dimension	d_{act}	2 per SUV	Continuous east–north velocity command in $[-1, 1]^2$, followed by circular projection to enforce the maximum-speed constraint.
Parallel environments	N_{env}	4	Number of vectorized training environments.
Rollout length	n_{steps}	2048	Number of steps collected per environment before each PPO update.
Mini-batch size	B	256	Mini-batch size used during each gradient update.
Optimization epochs	N_{epochs}	10	Number of passes over each rollout.
Learning rate	α	3×10^{-4}	Adam learning rate used for the actor and critic.
Discount factor	γ	0.99	Reward discount factor.
GAE parameter	λ_{GAE}	0.95	Generalized advantage estimation applied to the team-average reward.
Policy clipping range	ϵ	0.2	Clipping range of the PPO probability ratio.
Value clipping range	ϵ_v	0.2	Clipping range of the value-function update.
Entropy coefficient	c_{ent}	0.01	Entropy regularization coefficient.
Value-loss coefficient	c_v	0.5	Weight of the value-function loss.
Maximum gradient norm	$g_{\max}$	0.5	Global gradient clipping threshold.
Training steps	T	2×10^6	Nominal environment steps per training run.
Independent seeds	–	0–9	Ten random seeds for each selected surface or active-control configuration.
Surface training forecast-error radius	R_{train}	0–800 km	The candidate surface formation-center radial offset is sampled uniformly in [0, 800] km, with a uniformly sampled bearing. This models the assumed upper bound of TC track-forecast error; offshore-only rejection sampling is then applied as part of the safety layer.
Surface validation center offsets	R_{test}	100/200/400/600/800 km	Fixed surface formation-center offsets with random bearings, evaluated without retraining. These values represent increasing track-forecast displacement.
Underwater initialization radius	$R_{0,\text{u}}$	≤200 km	At the end time of each surface segment, the separate subsurface episode initializes four vehicles with radial offsets sampled uniformly in [0, 200] km and uniformly sampled bearings. Surface endpoint positions are not transferred. The 0–800 km forecast-error protocol is surface-only.
Decision interval	Δt	10 min	Environmental observation and action interval.
Surface formation radius	R_{ring}	100 km	Surface SUVs use equal angular spacing of $360°/N$ on a 100 km ring.
Surface fleet sizes	N	4/8/12/16	Number of SUVs in the final surface configurations.
Underwater fleet size	N	4	Number of SUVs in each active-IPPO, random-control, and passive-drift case.
Underwater depths below the sea surface	z	0/10/20/30/40/50/60 m	Seven ocean-current observation depths used in the final subsurface matrix.
Underwater methods	–	Active IPPO; random; drift	Three paired methods evaluated on the same episodes and deployments.
Underwater duration	H	20 h	Fixed simulation horizon for the final subsurface comparison; this is not an endurance specification.
Surface speed limits	$V^{\text{s}}_{\max}$	12 or 20 m/s	Two surface-speed scenarios.
Submerged speed limits	$V^{\text{u}}_{\max}$	1.5 or 2.0 m/s	Two resultant propulsion-speed limits after unit-disk action projection, used for the active-control comparison.
Compute device	–	CPU	All reported IPPO experiments were trained on CPU.

References

[1] T. Knutson, S. J. Camargo, J. C. L. Chan, et al., Tropical Cyclones and Climate Change Assessment: Part I: Detection and Attribution, Bulletin of the American Meteorological Society 100 (10) (2019) 1987–2007. doi:10.1175/BAMS-D-18-0189.1.

[2] T. Knutson, S. J. Camargo, J. C. L. Chan, et al., Tropical Cyclones and Climate Change Assessment: Part II: Projected Response to Anthropogenic Warming, Bulletin of the American Meteorological Society 101 (3) (2020) E303–E322. doi:10.1175/BAMS-D-18-0194.1.

[3] S. S. Chand, K. J. E. Walsh, S. J. Camargo, et al., Declining Tropical Cyclone Frequency under Global Warming, Nature Climate Change 12 (7) (2022) 655–661. doi:10.1038/s41558-022-01388-4.

[4] Y. Li, Y. Tang, S. Wang, et al., Recent Increases in Tropical Cyclone Rapid Intensification Events in Global Offshore Regions, Nature Communications 14 (1) (2023) 5167. doi:10.1038/s41467-023-40605-2.

[5] W. Mei, S.-P. Xie, Intensification of Landfalling Typhoons over the Northwest Pacific since the Late 1970s, Nature Geoscience 9 (10) (2016) 753–757. doi:10.1038/ngeo2792.

[6] L. Liu, Y. Wang, R. Zhan, et al., Increasing Destructive Potential of Landfalling Tropical Cyclones over China, Journal of Climate 33 (9) (2020) 3731–3743. doi:10.1175/JCLI-D-19-0451.1.

[7] K. Balaguru, C.-C. Chang, L. R. Leung, et al., A Global Increase in Nearshore Tropical Cyclone Intensification, Earth's Future 12 (5) (2024) e2023EF004230. doi:10.1029/2023EF004230.

[8] J. P. Kossin, K. A. Emanuel, S. J. Camargo, Past and Projected Changes in Western North Pacific Tropical Cyclone Exposure, Journal of Climate 29 (16) (2016) 5725–5739. doi:10.1175/JCLI-D-16-0076.1.

[9] J. P. Cangialosi, E. Blake, M. DeMaria, et al., Recent Progress in Tropical Cyclone Intensity Forecasting at the National Hurricane Center, Weather and Forecasting 35 (5) (2020) 1913–1922. doi:10.1175/WAF-D-20-0059.1.

[10] X. Huang, X. Peng, J. Fei, et al., Evaluation and Error Analysis of Official Tropical Cyclone Intensity Forecasts during 2005–2018 for the Western North Pacific, Journal of the Meteorological Society of Japan. Ser. II 99 (1) (2021) 139–163. doi:10.2151/jmsj.2021-008.

[11] H. Zhang, H. He, W.-Z. Zhang, et al., Upper Ocean Response to Tropical Cyclones: A Review, Geoscience Letters 8 (1) (2021) 1. doi:10.1186/s40562-020-00170-8.

[12] K. Balaguru, P. Chang, R. Saravanan, et al., Ocean Barrier Layers' Effect on Tropical Cyclone Intensification, Proceedings of the National Academy of Sciences 109 (36) (2012) 14343–14347. doi:10.1073/pnas.1201364109.

[13] M. A. Bourassa, T. Meissner, I. Cerovecki, et al., Remotely Sensed Winds and Wind Stresses for Marine Forecasting and Ocean Modeling, Frontiers in Marine Science 6 (2019) 443. doi:10.3389/fmars.2019.00443.

[14] B. W. Klotz, E. W. Uhlhorn, Improved Stepped Frequency Microwave Radiometer Tropical Cyclone Surface Winds in Heavy Precipitation, Journal of Atmospheric and Oceanic Technology 31 (11) (2014) 2392–2408. doi:10.1175/JTECH-D-14-00028.1.

[15] R. Domingues, A. Kuwano-Yoshida, P. Chardon-Maldonado, et al., Ocean Observations in Support of Studies and Forecasts of Tropical and Extratropical Cyclones, Frontiers in Marine Science 6 (2019) 446. doi:10.3389/fmars.2019.00446.

[16] J. D. Doyle, J. R. Moskaitis, J. W. Feldmeier, et al., A View of Tropical Cyclones from Above: The Tropical Cyclone Intensity Experiment, Bulletin of the American Meteorological Society 98 (10) (2017) 2113–2134. doi:10.1175/BAMS-D-16-0055.1.

[17] J. J. Cione, et al., Eye of the Storm: Observing Hurricanes with a Small Unmanned Aircraft System, Bulletin of the American Meteorological Society 101 (2) (2020) E186–E205. doi:10.1175/BAMS-D-19-0169.1.

[18] D. L. Rudnick, Ocean Research Enabled by Underwater Gliders, Annual Review of Marine Science 8 (1) (2016) 519–541. doi:10.1146/annurev-marine-122414-033913.

[19] J. Hwang, N. Bose, S. Fan, AUV Adaptive Sampling Methods: A Review, Applied Sciences 9 (15) (2019) 3145. doi:10.3390/app9153145.

[20] J. Dong, R. Domingues, G. Goni, et al., Impact of Assimilating Underwater Glider Data on Hurricane Gonzalo (2014) Forecasts, Weather and Forecasting 32 (3) (2017) 1143–1159. doi:10.1175/WAF-D-16-0182.1.

[21] G. R. Halliwell, G. J. Goni, M. F. Mehari, et al., OSSE Assessment of Underwater Glider Arrays to Improve Ocean Model Initialization for Tropical Cyclone Prediction, Journal of Atmospheric and Oceanic Technology 37 (3) (2020) 467–487. doi:10.1175/JTECH-D-18-0195.1.

[22] X. Chen, F. Zhang, Development of a Convection-Permitting Air–Sea-Coupled Ensemble Data Assimilation System for Tropical Cyclone Prediction, Journal of Advances in Modeling Earth Systems 11 (11) (2019) 3474–3496. doi:10.1029/2019MS001795.

[23] L. Phillipson, Y. Li, R. Toumi, Strongly Coupled Assimilation of a Hypothetical Ocean Current Observing Network within a Regional Ocean–Atmosphere Coupled Model: An OSSE Case Study of Typhoon Hato, Monthly Weather Review 149 (5) (2021) 1317–1336. doi:10.1175/MWR-D-20-0108.1.

[24] L. Liu, A. Mehra, D. Kleist, et al., Impact of Assimilating Satellite and Glider Observations on Hurricane Isaias (2020) Forecast Using Marine JEDI, Weather and Forecasting 38 (9) (2023) 1807–1826. doi:10.1175/WAF-D-22-0014.1.

[25] S. Chen, J. A. Cummings, J. M. Schmidt, et al., Targeted Ocean Sampling Guidance for Tropical Cyclones, Journal of Geophysical Research: Oceans 122 (5) (2017) 3505–3518. doi:10.1002/2017JC012727.

[26] M. Mu, F. Zhou, H. Wang, A Method for Identifying the Sensitive Areas in Targeted Observations for Tropical Cyclone Prediction: Conditional Nonlinear Optimal Perturbation, Monthly Weather Review 137 (5) (2009) 1623–1639. doi:10.1175/2008MWR2640.1.

[27] C.-C. Wu, J.-H. Chen, P.-H. Lin, et al., Targeted Observations of Tropical Cyclone Movement Based on the Adjoint-Derived Sensitivity Steering Vector, Journal of the Atmospheric Sciences 64 (7) (2007) 2611–2626. doi:10.1175/JAS3974.1.

[28] C. Dong, G. Wu, Y. Zhang, H. Zhang, Q. Wang, An Innovative Submersible Unmanned Vessel: Blue Whale, Ocean-Land-Atmosphere Research 4 (2025) 0102. doi:10.34133/olar.0102.

[29] Y. Tian, W. Li, Z. Wang, Development and Validation of a Wave-Propelled Semi-Submersible Unmanned Vehicle, Polish Maritime Research 31 (4) (2024) 31–42. doi:10.2478/pomr-2024-0048.

[30] H. Zhou, K. Liu, H. Xu, X. Feng, Experimentally Verified Depth Control of an Unmanned Semi-Submersible Vehicle, IEEE Access 7 (2019) 94254–94262. doi:10.1109/ACCESS.2019.2925149.

[31] H. Chen, J. Li, Y. Xuan, et al., First Rocketsonde Launched from an Unmanned Semi-Submersible Vehicle, Advances in Atmospheric Sciences 36 (4) (2019) 339–345. doi:10.1007/s00376-018-8249-5.

[32] P. E. Hart, N. J. Nilsson, B. Raphael, A Formal Basis for the Heuristic Determination of Minimum Cost Paths, IEEE Transactions on Systems Science and Cybernetics 4 (2) (1968) 100–107. doi:10.1109/TSSC.1968.300136.

[33] S. Koenig, M. Likhachev, D* Lite, in: Eighteenth National Conference on Artificial Intelligence, 2002, pp. 476–483.

[34] S. Karaman, E. Frazzoli, Sampling-Based Algorithms for Optimal Motion Planning, The International Journal of Robotics Research 30 (7) (2011) 846–894. doi:10.1177/0278364911406761.

[35] O. Khatib, Real-Time Obstacle Avoidance for Manipulators and Mobile Robots, The International Journal of Robotics Research 5 (1) (1986) 90–98. doi:10.1177/027836498600500106.

[36] D. Q. Mayne, J. B. Rawlings, C. V. Rao, et al., Constrained Model Predictive Control: Stability and Optimality, Automatica 36 (6) (2000) 789–814. doi:10.1016/S0005-1098(99)00214-9.

[37] M. Panda, B. Das, B. Subudhi, et al., A Comprehensive Review of Path Planning Algorithms for Autonomous Underwater Vehicles, International Journal of Automation and Computing 17 (3) (2020) 321–352. doi:10.1007/s11633-019-1204-9.

[38] P. F. Christiano, J. Leike, T. Brown, et al., Deep Reinforcement Learning from Human Preferences, in: Advances in Neural Information Processing Systems, Vol. 30, 2017, pp. 4299–4307.

[39] L. Ouyang, J. Wu, X. Jiang, et al., Training Language Models to Follow Instructions with Human Feedback, in: Advances in Neural Information Processing Systems, Vol. 35, 2022, pp. 27730–27744. doi:10.52202/068431-2011.

[40] D. Guo, D. Yang, H. Zhang, et al., DeepSeek-R1 Incentivizes Reasoning in LLMs through Reinforcement Learning, Nature 645 (8081) (2025) 633–638. doi:10.1038/s41586-025-09422-z.

[41] J. Hwangbo, J. Lee, A. Dosovitskiy, et al., Learning Agile and Dynamic Motor Skills for Legged Robots, Science Robotics 4 (26) (2019) eaau5872. doi:10.1126/scirobotics.aau5872.

[42] OpenAI: Marcin Andrychowicz, B. Baker, M. Chociej, et al., Learning Dexterous In-Hand Manipulation, The International Journal of Robotics Research 39 (1) (2020) 3–20. doi:10.1177/0278364919887447.

[43] E. Kaufmann, L. Bauersfeld, A. Loquercio, et al., Champion-Level Drone Racing Using Deep Reinforcement Learning, Nature 620 (7976) (2023) 982–987. doi:10.1038/s41586-023-06419-4.

[44] O. Vinyals, I. Babuschkin, W. M. Czarnecki, et al., Grandmaster Level in StarCraft II Using Multi-Agent Reinforcement Learning, Nature 575 (7782) (2019) 350–354. doi:10.1038/s41586-019-1724-z.

[45] C. Berner, G. Brockman, B. Chan, et al., Dota 2 with Large Scale Deep Reinforcement Learning (2019). arXiv:1912.06680, doi:10.48550/arXiv.1912.06680.

[46] C. S. de Witt, T. Gupta, D. Makoviichuk, et al., Is Independent Learning All You Need in the StarCraft Multi-Agent Challenge? (2020). arXiv:2011.09533, doi:10.48550/arXiv.2011.09533.

[47] C. Yu, A. Velu, E. Vinitsky, et al., The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games, in: Advances in Neural Information Processing Systems, Vol. 35, 2022, pp. 24611–24624. doi:10.52202/068431-1787.

[48] M. Samvelyan, T. Rashid, C. S. de Witt, et al., The StarCraft Multi-Agent Challenge, in: Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems, International Foundation for Autonomous Agents and Multiagent Systems, 2019, pp. 2186–2188. doi:10.65109/lvzz5205.

[49] E. Fiorelli, N. E. Leonard, P. Bhatta, et al., Multi-AUV Control and Adaptive Sampling in Monterey Bay, IEEE Journal of Oceanic Engineering 31 (4) (2006) 935–948. doi:10.1109/JOE.2006.880429.

[50] I. Masmitja, M. Martin, T. O'Reilly, et al., Dynamic Robotic Tracking of Underwater Targets Using Reinforcement Learning, Science Robotics 8 (80) (2023) eade7811. doi:10.1126/scirobotics.ade7811.

[51] S. Wang, C. Lin, G. Han, et al., Multi-AUV Cooperative Underwater Multi-Target Tracking Based on Dynamic-Switching-Enabled Multi-Agent Reinforcement Learning, IEEE Transactions on Mobile Computing 24 (5) (2025) 4296–4311. doi:10.1109/TMC.2024.3521889.

[52] Z. Zhao, Q. Hu, H. Feng, et al., A Cooperative Hunting Method for Multi-AUV Swarm in Underwater Weak Information Environment with Obstacles, Journal of Marine Science and Engineering 10 (9) (2022) 1266. doi:10.3390/jmse10091266.

[53] X. Li, P. Dong, H. Tao, et al., A Multi-AUV Cooperative Search Scheme Based on Acoustic-Optical Communication and Deep Reinforcement Learning, in: 2024 IEEE International Conference on High Performance Computing and Communications, 2024, pp. 785–790. doi:10.1109/HPCC64274.2024.00108.

[54] F. Qi, X. Ma, J. Yang, Tropical Cyclone Tracking for Autonomous Underwater Vehicles Based on Forecast Path Correction Model, Ocean Engineering 293 (2024) 116768. doi:10.1016/j.oceaneng.2024.116768.

[55] F. Qi, X. Ma, Y. Ding, J. Yang, Unmanned Active Observation Planning of Near-Surface Ocean under Tropical Cyclone Condition, Journal of Ocean Engineering and Science 11 (6) (2026) 2077–2088. doi:10.1016/j.joes.2026.05.012.

[56] J. Schulman, F. Wolski, P. Dhariwal, et al., Proximal Policy Optimization Algorithms (2017). arXiv:1707.06347, doi:10.48550/arXiv.1707.06347.

[57] H. Hersbach, B. Bell, P. Berrisford, et al., The ERA5 Global Reanalysis, Quarterly Journal of the Royal Meteorological Society 146 (730) (2020) 1999–2049. doi:10.1002/qj.3803.

[58] E. P. Chassignet, H. E. Hurlburt, O. M. Smedstad, et al., The HYCOM (HYbrid Coordinate Ocean Model) Data Assimilative System, Journal of Marine Systems 65 (1–4) (2007) 60–83. doi:10.1016/j.jmarsys.2005.09.016.

[59] K. R. Knapp, M. C. Kruk, D. H. Levinson, et al., The International Best Track Archive for Climate Stewardship (IBTrACS): Unifying Tropical Cyclone Data, Bulletin of the American Meteorological Society 91 (3) (2010) 363–376. doi:10.1175/2009BAMS2755.1.